

THE DUAL APPROACH TO RECURSIVE OPTIMIZATION: THEORY AND EXAMPLES

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We develop a recursive dual method for solving dynamic economic problems. The method uses a Lagrangian to pair a dynamic recursive economic problem with a dual problem. We show that such dual problems can be recursively decomposed with costates (i.e., Lagrange multipliers on laws of motion) functioning as state variables. In dynamic contracting and policy settings, the method often replaces an endogenous state space of forward-looking utilities with an exogenously given state space of costates. We provide a principle of optimality for dual problems and give conditions under which the dual Bellman operator is a contraction with the optimal dual value function its unique fixed point. We relate economic problems to their duals, address computational issues, and give examples.

KEYWORDS: Dynamic contracts, duality, dynamic programming, contraction mapping.

1. INTRODUCTION

MANY DYNAMIC ECONOMIC RELATIONSHIPS can be formulated as contracting or planning problems subject to limited commitment or private information frictions. These problems often admit recursive formulations in which utility promises to agents serve as state variables. Under such formulations, past and future promises enter into and ensure consistency of the problems' incentive constraints across periods.¹ However, whereas in standard problems the state space is a primitive, in these problems it is a set of incentive-feasible utility promises that is not known a priori. Consequently, numerical implementation of this recursive approach requires prior (or joint) recovery of an endogenous state space. In addition, while the associated optimal value functions (and sometimes state spaces) are fixed points of Bellman-like operators, these operators are rarely contractions.² The goal of this paper is to augment the economic practitioner's tool kit with a recursive dual method for characterizing and solving such problems.

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¹Examples are given in the paper. For further examples, see [Ljungqvist and Sargent \(2012\)](#).

²In addition to contracting problems, such difficulties emerge in other settings featuring forward-looking state variables, that is, settings in which the state variable is defined as a function of future actions and shocks and the set of such state variables is not given a priori. Examples include portfolio problems in which the agent faces a terminal bankruptcy constraint or monetary policy problems in which the price level is given by a forward-looking New Keynesian Phillips curve.

Under this method, a dynamic economic optimization is replaced by a closely related dual problem. The latter is then formulated recursively with its value and policy functions defined on a state space of costates. In contracting and planning problems, these costates are typically multipliers on laws of motion for promises and have natural economic interpretations as Pareto weights. We show that for a large class of problems, the dual problem's (co)state space is known a priori and does not need to be separately computed. The recursive dual value function is a fixed point of a Bellman operator. We give conditions under which this Bellman operator is a contraction. We discuss numerical implementation of the recursive dual method and apply it to a limited commitment problem with recursive (Epstein–Zin–Kreps–Porteus) preferences. Since the recursive dual method solves the dual rather than the original problem, it is important to know when the optimal payoffs and solutions from the dual and the original problem coincide. We give sufficient conditions for this. For problems in which these conditions are not satisfied, we provide a numerical check for original problem optimality.

To introduce and motivate the recursive dual method, we use a risk sharing problem with limited commitment and recursive preferences.³ In this problem, a planner seeks to maximize a Pareto weighted sum of agent utilities subject to resource and no default constraints. The latter require that each agent's utility remains above the utility from an outside default option. This problem can be formulated recursively with current shocks and utility promises to agents as state variables. The corresponding state space is then a set of shocks and incentive-feasible utility promises. This is not known a priori and must be computed jointly with the problem's value and policy functions or in a prior step. The analogue of the Bellman operator for this problem, while monotone, is not a contraction. In contrast, the recursive dual problem uses an exogenously given state space of costates (i.e., Lagrange multipliers on the law of motion for utility promises or, in more economic terms, Pareto weights) and has a Bellman operator that is a contraction (on an appropriate set of candidate value functions).

Derivation of the recursive dual problem in this and other settings is in two steps. In the first step, a non-recursive dual problem is defined via a Lagrangian. This problem involves an outer minimization over the Lagrangian's multipliers and costates and an inner maximization over consumption allocations. The recursive structure of the original problem is inherited by the dual with the costates serving as dual state variables that are passed between periods. Intuitively, agents with high current outside options (and binding default constraints) are assigned higher next period costates (or Pareto weights) to ensure that they receive additional utility in the future and do not default today. The dual value function (on a domain of costates) is a fixed point of a dual Bellman operator. This operator updates candidate value functions via an outer minimization over current constraint multipliers and next period costates and an inner maximization over current consumption and next period utility promises. The inner maximizations are simple and separable. They often have analytical solutions in which case application of the Bellman reduces to a minimization of an indirect value function over Lagrange multipliers.

The risk sharing problem's optimal dual value function has several convenient properties. It is convex and positively homogenous in costates and finite on $\mathcal{S} \times \mathbb{R}_+^{n_I}$, where $\mathcal{S}$ is the set of shocks and $\mathbb{R}_+^{n_I}$ the set of costates for the n_I agents. The positive homogeneity property implies that the dual value function is fully defined on the compact set $\mathcal{S} \times \mathcal{C}_+$, where $\mathcal{C}_+$ is the intersection of the unit sphere with $\mathbb{R}_+^{n_I}$. Thus, the dual state space

³This problem has been widely analyzed with standard expected utility preferences; see, for example, Kocherlakota (1996).

is simple and is immediately given. The dual Bellman operator is a contraction (on an appropriately defined interval of functions and with respect to an alternative Thompson metric). Hence, it is natural to apply value iteration to calculate the optimal value function. We do so, calculating solutions to two and three agent problems. The strict concavity of the objective and constraint functions combined with the structure of the constraint set ensures that the allocation computed via the recursive dual approach is (up to numerical error) a solution to the original problem.

The remainder of the paper extends and rigorously proves, in a general environment, results discussed in the limited commitment setting. Specifically, it considers a family of dynamic recursive optimizations that encompasses many economic applications. The objective and constraints of these optimizations are functions of current actions and recursively evolving state variables. In the context of particular applications, the state variables have interpretations as capital, utility promises, or inflation. They may be backward-looking (i.e., functions of past actions and shocks and an initial condition such as physical capital) or forward-looking (i.e., functions of current and future actions and shocks as the utility promises were in the limited commitment case). As before, a dual problem is associated with these optimizations via a Lagrangian. For the recursive dual formulation to accommodate laws of motion for state variables that are nonlinear in states, the Lagrangian must explicitly include these laws of motion along with other constraints.⁴ In this general setting, we formally develop the computational merits of the recursive dual approach. We give conditions for the dual Bellman operator to be a contraction on a space of convex, positively homogenous, and readily approximable functions. As in the limited commitment case, the (co)state space may be restricted to the corresponding unit sphere or its intersection with the nonnegative orthant. We describe how, in the general case, the dual Bellman again involves minimizations over multipliers that coordinate low-dimensional and sometimes analytically solvable maximizations. Finally, we provide theoretical conditions for the equality of optimal dual and original problem payoffs and stronger conditions for equality of solutions. For situations in which these conditions fail, we provide a numerical ex post check of dual-original problem equivalence.

Establishing that the dual Bellman operator is a contraction presents special challenges. The recursive dual features an unbounded value function and an unbounded constraint correspondence. For problems with unbounded value functions, a common procedure, following [Wessels \(1977\)](#), is to show that there is a set of functions closed and bounded with respect to a more permissive weighted sup norm that contains the optimal value function and on which the Bellman is a contractive self-map. However, this approach requires that the continuation state variables and, hence, the continuation value function cannot vary “too much” on the graph of the constraint correspondence. Since the dual Bellman operator permits the choice of multipliers from an unbounded set, this condition is only guaranteed in the dual setting if additional non-binding constraints on multipliers are found. Instead, we show that the Bellman is contractive with respect to an alternative metric on a space of functions sandwiched between two (unbounded) functions. We show through examples that such bounding functions are often available.

The paper proceeds as follows. After a brief literature review, [Section 2](#) uses a limited commitment model with recursive preferences to introduce the recursive dual approach. [Section 3](#) introduces a general class of stochastic, infinite-horizon decision problems. In

⁴If laws of motion are linear in states, then state variables can be substituted from the problem via iteration on their laws of motion. Both states and laws may then be excluded from the Lagrangian.

Section 4, a decision maker's problem from this class is paired with its dual and a recursive formulation of the latter obtained. A Bellman-type principle of optimality for the dual problem is established. Section 5 gives a contraction result for recursive dual problems. Section 6 relates values and solutions to primal and dual problems. Section 7 concludes. The published appendix contains proofs. Details of numerical implementation and additional calculations from the limited commitment case are contained in the Supplemental Material (Pavoni, Sleet and Messner (2018)).

Literature

Our method is related to, but distinct from, that of Marcet and Marimon (2011). These authors proposed solving dynamic optimizations by recursively decomposing a saddle point operation. They restricted attention to concave problems with constraints (including laws of motion) that are linear in forward-looking state variables. They substituted forward-looking states out of the problem using their laws of motion and absorbed a subset of constraints into a Lagrangian. Laws of motion for backward-looking primal states (e.g., capital) are left as explicit restrictions. They then recursively decomposed a saddle point of this Lagrangian (on the constraint set defined by the backward-looking laws of motion). This approach requires that a saddle point exists in all subproblems after all histories.

Our approach cleanly separates dualization of the problem from recursive decomposition of the dual. We show that the latter is available under weak separability conditions, much weaker than those assumed by Marcet and Marimon (2011). Theoretical sufficient conditions for equality of optimal dual and original problem payoffs and solutions are stronger and more restrictive. Even so, several of Marcet and Marimon's restrictions can be dispensed with. Specifically, the requirements that constraints are linear in forward-looking state variables and that *every* continuation problem has a saddle can be dropped. Moreover, when our theoretical conditions are not satisfied, we propose a numerical procedure for checking optimality of a dual solution. We also go beyond Marcet and Marimon (2011) in proving that the dual Bellman operator is a contraction and in describing how to numerically implement the recursive dual approach.⁵

Messner, Pavoni, and Sleet (2012) considered the relationship between primal (i.e., those obtained from standard recursive formulations) and dual Bellman operators. They restricted attention to concave problems without backward-looking state variables and with laws of motion that are linear in forward-looking ones. Thus, their setting is less general than the present one: it excludes problems such as default with capital accumulation, risk sharing with non-expected utility, and optimal monetary policy. In addition, they did not provide contraction results or a numerical implementation. In an important extension of Marcet and Marimon (2011), Cole and Kubler (2012) showed how recursive methods using dual variables can be adapted to give sufficient conditions for an optimal solution under weak concavity. Their method expands the state space to incorporate realizations of an end-of-period lottery over the extreme points of flat regions of the continuation value function. This approach permits optimal policies to be isolated in the convexifications of

⁵For some problems, Marcet and Marimon's (2011) recursive saddle Bellman operator is available and resembles our dual Bellman. However, in these cases, our Bellman operator implements a fairly straightforward inf-sup operation, whereas theirs involves a more difficult saddle point operation. In other cases, the recursive saddle Bellman operator is not available, or is available but is quite different from ours. In particular, several of the examples considered in this paper and its appendices either cannot be handled by Marcet and Marimon (2011)'s formulation or would be handled differently.

otherwise non-concave problems. We leave integration of Cole and Kubler's (2012) insights into our framework for future work. In a continuous time setting, Miao and Zhang (2015) solved a limited commitment problem using a recursive dual approach. In this specific setting, Miao and Zhang (2015) obtained a sharp characterization of a solution to the problem. In contrast, we consider discrete time settings (where the mathematics is quite different) and develop a method applicable to a range of problems.

Abreu, Pearce, and Stacchetti (1990) provided a recursive characterization of payoff sets in repeated games. Applied to our setting, their formulation gives an alternative recursive primal approach to our recursive dual one. Numerical implementations of this approach have been provided by Chang (1998) and Judd, Yeltekin, and Conklin (2003). A recent alternative and highly promising approach was provided by Judd, Renner, Scheidegger, and Yeltekin (2017), who used a combination of penalty functions and adaptive splines to solve a non-concave dynamic optimal tax problem with an endogenous state space. Many economists have adopted recursive formulations that replace or supplement standard "primal" state variables with "dual" ones. Examples include Kehoe and Perri (2002), Marimon and Quadrini (2006), Acemoglu, Golosov, and Tsyvinski (2010), Chien, Cole, and Lustig (2011), and Aiyagari, Marcet, Sargent, and Seppälä (2002). Despite their widespread use, thorough analysis of these methods is limited and their application has often been ad hoc.

2. THE DUAL RECURSIVE METHOD: AN EXAMPLE

This section uses a model of risk sharing with limited commitment and recursive preferences to introduce the dual recursive method. Limited commitment models with standard expected utility preferences have found application in macroeconomics, finance, and development.⁶ The extension to recursive preferences illustrates the flexibility of our method and is of independent interest. To maximize accessibility for practitioners, our tone is heuristic throughout this section. Formal derivations of results in a more general setting are given in later sections.

2.1. *A Limited Commitment Setting*

A group of agents indexed by $i \in \mathcal{I} = \{1, \dots, n_I\}$ and subject to shocks to their endowment and the value of their outside option seek to insure risk. At each date, these agents have the opportunity to quit the insurance arrangement and take their default outside options. Their inability to commit not to do this limits risk sharing.

Let $s \in \mathcal{S} = \{1, \dots, n_S\}$ be a current shock and $s^t = (s_1, \dots, s_t) \in \mathcal{S}^t$ be a shock history of length $t = 1, 2, \dots, \infty$. Shocks are assumed to evolve according to a Markov process with transition matrix π . Let $\pi(s'|s)$ denote the probability of transitioning from shock s in the current period to s' in the next period and let $\pi^t(s'|s_0)$ denote the induced probability of shock history s^t conditional on initial shock s_0 . Shocks impact the joint endowment of the

⁶Economists have studied risk sharing without commitment amongst family members, partners in businesses, villagers in developing economies, countries in sovereign debt markets, and so on. Thomas and Worrall (1988) and Kocherlakota (1996) provided theoretical analyses of economies with two agents. Ligon, Thomas, and Worrall (2002) extended the analysis to a setting with finite (and more than two) agents and provided an empirical application. Numerical analysis, however, has been limited to games played by two or a continuum of agents. The latter reduce to one sided commitment problems at endogenous market clearing shadow prices. We are not aware of any analysis of the risk sharing model with limited commitment that uses preferences different from the standard expected utility.

agents and the default outside utility option obtained by an agent on leaving the group. Let $\gamma : \mathcal{S} \rightarrow \mathbb{R}_{++}$ give the endowment of the group after each shock and let $w : \mathcal{S} \rightarrow \mathbb{R}^{\mathcal{I}}$, $w(s) = \{w^i(s)\}_{i \in \mathcal{I}}$, give agents' default outside utility options.

Agents derive utility from consumption. Let $\mathcal{A} := \times_{i \in \mathcal{I}} [\underline{a}^i, \bar{a}^i] \subset \mathbb{R}_+^{\mathcal{I}}$ be a set of possible per period consumption levels for the group. Agents' choices at each history are collected into a *consumption plan*: $\mathbf{a} = \{a_t\}_{t=0}^\infty$, with $a_0 \in \mathcal{A}$ and for $t \geq 1$, $a_t : \mathcal{S}^t \rightarrow \mathcal{A}$. The i th component of a_t represents the consumption of the i th agent at date t and is denoted a_t^i ; the s^t -continuation of a consumption plan $\mathbf{a}$ is denoted $\mathbf{a}|s^t = \{a_{t+\tau}(s^t, \cdot)\}_{\tau=0}^\infty$.⁷ Denote the set of consumption plans by $\mathbf{A}$. The payoff to the i th agent from consumption plan $\mathbf{a}$ given initial shock s_0 is denoted $V^i(s_0, \mathbf{a})$, where V^i satisfies the Kreps–Porteus/Epstein–Zin recursion

$$V^i(s_t, \mathbf{a}|s^t) = \left[(1 - \delta)[a_t^i(s^t)]^{1-\sigma} + \delta \left\{ \sum_{s' \in \mathcal{S}} [V^i(s', \mathbf{a}|(s^t, s'))]^{1-\rho} \pi(s'|s) \right\}^{\frac{1-\rho}{1-\sigma}} \right]^{\frac{1}{1-\sigma}}, \quad (\text{U})$$

with $0 < \sigma$, $0 < \rho$, and $0 < \delta < 1$. Here $1/\sigma$ is the intertemporal elasticity of substitution (IES), ρ the coefficient of relative risk aversion (CRRA), and δ the intertemporal discount factor. The compactness of $\mathcal{A}$ ensures that $V = \{V^i\}_{i \in \mathcal{I}}$ takes its values in a bounded set $\mathcal{V}$.

The planner attaches Pareto weights $\lambda = \{\lambda^i\}_{i \in \mathcal{I}} \geq 0$ to the agents' date zero payoffs and values consumption plans according to

$$\sum_{i \in \mathcal{I}} \lambda^i V^i(s_0, \mathbf{a}). \quad (\text{O})$$

To be feasible a plan $\mathbf{a}$ must satisfy, for all $t \geq 0$, $s^t \in \mathcal{S}^t$, the *no default constraints*,

$$\text{for each } i \in \mathcal{I}, \quad V^i(s_t, \mathbf{a}|s^t) - w^i(s_t) \geq 0, \quad (\text{D})$$

and the *resource constraints*,

$$\gamma(s_t) - \sum_{i \in \mathcal{I}} a_t^i(s^t) \geq 0. \quad (\text{R})$$

The no default constraints ensure that each agent is better off remaining in the risk sharing arrangement than taking her default outside option. The planner must trade off her desire to smooth an agent's share of the endowment over time against the need to respond to large outside option shocks. The latter require raising a given agent's consumption in order to satisfy the no default constraint. The planner's problem at initial shock s_0 and Pareto weighting λ is

$$P_0^* = \sup_{\mathbf{a} \in \mathbf{A}} \sum_{i \in \mathcal{I}} \lambda^i V^i(s_0, \mathbf{a}) \quad \text{s.t. } \forall t \geq 0, s^t \in \mathcal{S}^t, (\text{D}) \text{ and } (\text{R}). \quad (\text{PP})$$

2.2. The Dual Problem and the Recursive Dual Approach

Given a consumption plan $\mathbf{a}$, define a corresponding *promise plan* $\mathbf{v} = \{v_t\}_{t=0}^\infty$ according to $v_0 := V(s_0, \mathbf{a}) = \{V^i(s_0, \mathbf{a})\}_{i \in \mathcal{I}} \in \mathcal{V}$, and for all $t \geq 1$, $s^t \in \mathcal{S}^t$,

$$v_t(s^t) := V(s_t, \mathbf{a}|s^t) = \{V^i(s_t, \mathbf{a}|s^t)\}_{i \in \mathcal{I}} \in \mathcal{V}. \quad (1)$$

⁷With $\mathbf{a}|s^0 := \mathbf{a}$ and $a_0(s^0) := a_0$.

Let $\mathbf{p} := (\mathbf{a}, \mathbf{v})$ denote a plan for both consumption and utility promises. It is readily verified that $\mathbf{p}$ is feasible if and only if it is consistent with the law of motion for utility promises (U) and the no default (D) and resource (R) constraints, that is, if $\forall t \geq 0$, $s^t \in \mathcal{S}^t$, $z_t^v(\mathbf{p})(s^t) = z^v(v_t(s^t), s_t, a_t(s^t), v_{t+1}(s^t, \cdot)) = 0$, where z^v evaluates the current utility promise law of motion:

$$z^v(v, s, a, v') := \left\{ \left[(1 - \delta)[a^i(s)]^{1-\sigma} + \delta \left\{ \sum_{s' \in \mathcal{S}} [v'^i(s')]^{1-\rho} \pi(s'|s) \right\}^{\frac{1-\sigma}{1-\rho}} \right]^{\frac{1}{1-\sigma}} - v^i(s) \right\}_{i \in \mathcal{I}},$$

and $z_t^h(\mathbf{p})(s^t) = z^h(s_t, a_t(s^t), v_{t+1}(s^t, \cdot)) \geq 0$, where z^h evaluates the current no default and resource constraint functions:

$$z^h(s, a, v') := \left(\left\{ \left[(1 - \delta)[a^i]^{1-\sigma} + \delta \left\{ \sum_{s' \in \mathcal{S}} [v'^i(s')]^{1-\rho} \pi(s'|s) \right\}^{\frac{1-\sigma}{1-\rho}} \right]^{\frac{1}{1-\sigma}} - w^i(s) \right\}_{i \in \mathcal{I}} \right. \\ \left. \gamma(s) - \sum_{i \in \mathcal{I}} a^i(s) \right).$$

Let $\mathbf{q} := (\mathbf{m}, \mathbf{y})$ be a multiplier plan, where $\mathbf{m} = \{m_t\}_{t=0}^\infty$ are nonnegative multipliers on the no default and resource constraints and $\mathbf{y} = \{y_t\}_{t=0}^\infty$ are nonnegative multipliers on the law of motion for promises, that is, are utility promise costates or, in economic terms, are endogenously evolving Pareto weights. Define the Lagrangian:

$$\mathcal{L}(\mathbf{p}, \mathbf{q}) := \lambda \cdot v_0 + \sum_{t=0}^{\infty} \sum_{s^t} \delta^t \{ y_t(s^t) \cdot z_t^v(\mathbf{p})(s^t) + m_t(s^t) \cdot z_t^h(\mathbf{p})(s^t) \} \pi^t(s^t | s_0).$$

The planner's problem may then be re-expressed as the *primal* sup-inf problem:⁸

$$P_0^* := \sup_{\mathbf{p}} \inf_{\mathbf{q}} \mathcal{L}(\mathbf{p}, \mathbf{q}). \quad (\text{SI}_0)$$

Problem (SI₀) delivers both the optimal value P_0^* and the set of solutions to (PP). Intuitively, the inner minimization over multiplier plans “infinitely penalizes” and, hence, enforces the problem's constraints. The planner's *dual problem* interchanges the inf and sup operations:

$$D_0^* := \inf_{\mathbf{q}} \sup_{\mathbf{p}} \mathcal{L}(\mathbf{p}, \mathbf{q}). \quad (\text{IS}_0)$$

Although problems (SI₀) and (IS₀) superficially resemble each other, they are in fact different and it is the latter that delivers our desired recursive formulation. Towards this formulation, define the *dual value function* D^* for all possible initial shocks $s_0 \in \mathcal{S}$ and

⁸The domain of the Lagrangian is the product of a space of plans $\mathbf{P}$ (inclusive of promises) and nonnegative multipliers $\mathbf{Q}$ (inclusive of costates on promises). Precise definitions are given in later sections. Note that although the law of motion for promises is an equality constraint, the monotonicity of the problem in these promises permits the relaxation of this constraint to an inequality. Consequently, the assumption that costates are nonnegative is unrestrictive.

weightings of initial utilities $y_0 \in \mathbb{R}_+^{n_I}$:

$$D^*(s_0, y_0) := \inf_q \sup_p y_0 \cdot v_0 + \sum_{t=0}^{\infty} \sum_{s^t} \delta^t \{ y_t(s^t) \cdot z_t^v(p)(s^t) + m_t(s^t) \cdot z_t^h(p)(s^t) \} \pi^t(s^t | \tilde{s}_0). \quad (2)$$

Note that $D_0^* = D^*(s_0, \lambda)$, that is, the optimal dual value is the value of D^* at the specific shock s_0 and the Pareto weighting λ . The recursive dual formulation of (IS₀) decomposes (2) into subproblems linked by shocks and costates on utility promises. Since the latter weight agent payoffs, it is natural to interpret them as endogenously evolving Pareto weights.

The recursive formulation is conveniently expressed in “time invariant” notation. Let $y \in \mathbb{R}_+^{n_I}$ be a costate for the current period and let $q = (m, y')$ be a *current dual choice*, where $m \in \mathbb{R}_+^{n_I+1}$ is a multiplier on the current constraints $z^h(s, p) \geq 0$ and $y' \in \mathbb{R}_+^{n_I \times n_S}$ is a state-contingent vector of costates for the next period. Let $p = (a, v')$ denote a *current primal choice*, where $a \in \mathcal{A}$ is current consumption and $v' \in \mathcal{V}^{n_S}$ is a state-contingent vector of continuation utility promises. The following dynamic programming result obtains.⁹

RESULT 1—Bellman: *The dual function D^* satisfies the Bellman equation:*

$$\forall (s, y) \in \mathcal{S} \times \mathbb{R}_+^{n_I}, \quad D^*(s, y) = \inf_q \sup_p J(s, y; q, p) + \delta \sum_{s' \in \mathcal{S}} D^*(s', y'(s')) \pi(s' | s), \quad (3)$$

where the current dual function J is given by

$$J(s, y; q, p) := y \cdot \underbrace{\left\{ \left[(1 - \delta)[a^i]^{1-\sigma} + \delta \left\{ \sum_{s' \in \mathcal{S}} [v'^i(s')]^{1-\rho} \pi(s' | s) \right\}^{\frac{1-\sigma}{1-\rho}} \right]^{\frac{1}{1-\sigma}} \right\}}_{\text{shadow value of utility today}} \Bigg\}_{i \in \mathcal{I}} \quad (4)$$

$$+ \underbrace{m \cdot z^h(s, p)}_{\text{shadow value of relaxing constraints}} - \underbrace{\delta \sum_{s' \in \mathcal{S}} y'(s') \cdot v'(s') \pi(s' | s)}_{\text{shadow cost of future promises}}.$$

The recursive dual problem (3) incorporates an inf-sup operation over current dual q and primal p choices. Its objective is the sum of a *current dual function* J and a discounted expected (*continuation*) *dual function* D^* . The current dual function J (defined in (4)) describes the shadow value of delivering utility to the agents inclusive of the benefit from relaxing the no default and resource constraints and net of the shadow cost of future utility promises. Heuristically, if agent i 's outside option $w^i(s)$ is large, the value of the element in $z^h(s, p)$ corresponding to this agent's no default constraint is reduced. The interaction of the inf and sup operations in (3) then lead to the choice of a higher multiplier m^i on the i th agent's no default constraint, higher continuation promises for agent i , $v'^i(\cdot)$, and higher costate $y'^i(\cdot)$ choices. The latter transmit the need for higher agent i utility to the future period. As noted, they function as endogenously evolving Pareto weights.

⁹The proof is a special case of Proposition 3 in Section 4.

Let $\mathcal{G}$ denote a candidate set of dual value functions containing D^* . The *dual Bellman operator* $\mathcal{B}$ implied by (3) is given by, for $D \in \mathcal{G}$ and each $(s, y) \in \mathcal{S} \times \mathbb{R}_+^{n_I}$,

$$\mathcal{B}(D)(s, y) = \inf_q \sup_p J(s, y; q, p) + \delta \sum_{s' \in \mathcal{S}} D(s', y'(s')) \pi(s'|s). \quad (5)$$

Thus, from (3), D^* is a fixed point of a Bellman operator: $D^* = \mathcal{B}(D^*)$. Plans for consumption (and multipliers) that solve the dual problem (IS₀) may be recovered directly from the policy correspondences associated with $\mathcal{B}(D^*)$. Further, such consumption plans also attain the optimum in the original planner's problem (PP).¹⁰ This recursive dual formulation has several convenient features from the point of view of computation.

1. *Contractivity.* Subject to the existence of certain bounding functions that define an appropriate domain $\mathcal{G}$ for the dual Bellman operator $\mathcal{B}$, this operator is a contraction.¹¹ $\mathcal{B}$ then has a unique fixed point, convergence of a value iteration from any D_0 in $\mathcal{G}$ to D^* is ensured, and both error bounds and rates of convergence are available.

2. *The state space is simple and exogenously given.* The value function D^* is real-valued at all $(s, y) \in \mathcal{S} \times \mathbb{R}_+^{n_I}$.¹² Moreover, D^* is positively homogenous, implying that it is fully defined on $\mathcal{S} \times \mathcal{C}_+$, where $\mathcal{C}_+$ is the intersection of the unit sphere with $\mathbb{R}_+^{n_I}$. Thus, the dual value function is fully determined on a simple compact set that can serve as the dual state space. This contrasts with the situation under recursive approaches that use utility promises to keep track of histories. Under these approaches, the state space is an endogenously determined set of promises.

3. *The inf sup operation defining $\mathcal{B}$ may be reduced to an inf operation.* Additive separability in the current dual function with respect to the elements of the primal choice p can be exploited to reduce the inner sup operation to a collection of simpler, parallelizable sup operations. Sometimes these simpler sup operations have analytical solutions that can be substituted into J , avoiding the need for any numerical maximization and reducing the application of $\mathcal{B}$ to a collection of minimizations over dual variables only. In the current setting, agent utilities may be (monotonically) transformed by applying the function $\tilde{v} := \frac{v^{1-\sigma}}{1-\sigma}$. The planner's problem can then be re-expressed in terms of transformed utilities. The law of motion for such utilities is¹³

$$\tilde{v}^i = \frac{1-\delta}{1-\sigma} [a^i]^{1-\sigma} + \frac{\delta}{1-\sigma} \left\{ \sum_{s' \in \mathcal{S}} [(1-\sigma)\tilde{v}^{i,s'}(s')]^{\frac{1-\rho}{1-\sigma}} \pi(s'|s) \right\}^{\frac{1-\sigma}{1-\rho}}. \quad (6)$$

The (time) additive separability of (6) may be exploited to re-express the indirect current dual objective $J^*(s, y; q) := \sup_{p \in \mathcal{A} \times \mathcal{V}^{n_S}} J(s, y; q, p)$ as

$$J^*(s, y; q) = m^{n_I+1} \gamma(s) - \sum_{i \in \mathcal{I}} m^i w^i(s) + \sup_{\mathcal{A}} \sum_{i \in \mathcal{I}} \left\{ \frac{1-\delta}{1-\sigma} (y^i + m^i) [a^i]^{1-\sigma} - m^{n_I+1} a^i \right\}$$

¹⁰These results follow from Propositions 4, 7, and 8 below.

¹¹This result is proved in a general setting in Theorem 2. The proof specializes to the limited commitment setting. In the Supplemental Material, bounding functions for this setting are derived.

¹²We prove this in a general setting in Proposition 2 below.

¹³To ensure that the transformed problem is concave, the restriction $\rho \geq \sigma$ is needed. To ensure that agent payoffs are bounded, if $\sigma < 1$, consumptions must be bounded below by an arbitrarily small $\underline{a} > 0$. In general, no default constraints will ensure that this last restriction is non-binding.

$$\begin{aligned}
& + \sup_{v^ns} \sum_{i \in \mathcal{I}} \left\{ \frac{\delta}{1-\sigma} (y^i + m^i) \left\{ \sum_s [(1-\sigma) \tilde{v}'^i(s')]^{\frac{1-\rho}{1-\sigma}} \pi(s'|s) \right\}^{\frac{1-\sigma}{1-\rho}} \right. \\
& \left. - \delta \sum_s y'^i(s') \tilde{v}'^i(s') \pi(s'|s) \right\}.
\end{aligned} \tag{7}$$

The component suprema in (7) are simple and easily calculated. When $(s, y; q)$ are such that the choices of a and v' are interior, then analytic solutions are available, the value of J^* is

$$\begin{aligned}
J^*(s, y; q) &= \frac{\sigma}{1-\sigma} \sum_{i \in \mathcal{I}} [(1-\delta)(y^i + m^i)]^{\frac{1}{\sigma}} (m^{n_I+1})^{\frac{\sigma-1}{\sigma}} \\
&\quad - \sum_{i \in \mathcal{I}} m^i w^i(s) + m^{n_I+1} \gamma(s),
\end{aligned} \tag{8}$$

and the co-states satisfy

$$\forall i \in \mathcal{I}, \quad y^i + m^i = \left\{ \sum_{s' \in \mathcal{S}} \{y'^i(s')\}^{\frac{1-\rho}{\sigma-\rho}} \pi(s'|s) \right\}^{\frac{\sigma-\rho}{1-\rho}}.$$

For these (s, y, q) values, no explicit inner maximization is necessary.

2.3. Numerical Results

This section reports numerical results from a two agent version of the problem.¹⁴ Its goal is to highlight qualitative properties of the model.

Given Result 1, approximations to D^* and the associated policy functions are obtained via value iteration.¹⁵ The preference parameters δ , σ , and ρ are set to 0.8, 1.5, and 5. Three shock values are assumed: $s \in \{1, 2, 3\}$. If $s = 1$ or 2, then agent s 's outside option is set to a high value, while the other agent's outside option is set to a low value.¹⁶ In the the third shock state, both outside options are set to the low value. The aggregate endowment is held constant at 1.2. The Markov transition matrix takes values of 0.9 on its leading diagonal, while off-diagonal elements are set to 0.05.

Figure 1 plots policy functions and simulations and, hence, illustrates the structure of the solution to the problem. As is well known from prior studies of limited commitment economies, if an agent has a low current utility promise and a high outside option, then her no default constraint binds and her current and future utilities must be increased. In the dual formulation, a low utility promise corresponds to a low current costate (or Pareto weight), while a binding no default constraint implies a positive value for the corresponding multiplier and an increased continuation costate. The latter transmits the need for a higher agent continuation utility to the future. Consequently, combinations of low current costate values and high outside options for an agent imply positive no default multipliers and increases in future costates. These features are apparent in Figure 1.

¹⁴Implications of the policy functions for the dynamics of the optimal allocation are most easily deduced in the two agent case. Results from a three agent version are given in Supplemental Material Appendix B.3.

¹⁵Details of the numerical implementation are given in Supplemental Material Appendix B.

¹⁶The high value equals the utility from receiving 52% of the aggregate endowment in each period; the low value is that from receiving 25% of the aggregate endowment in each period.

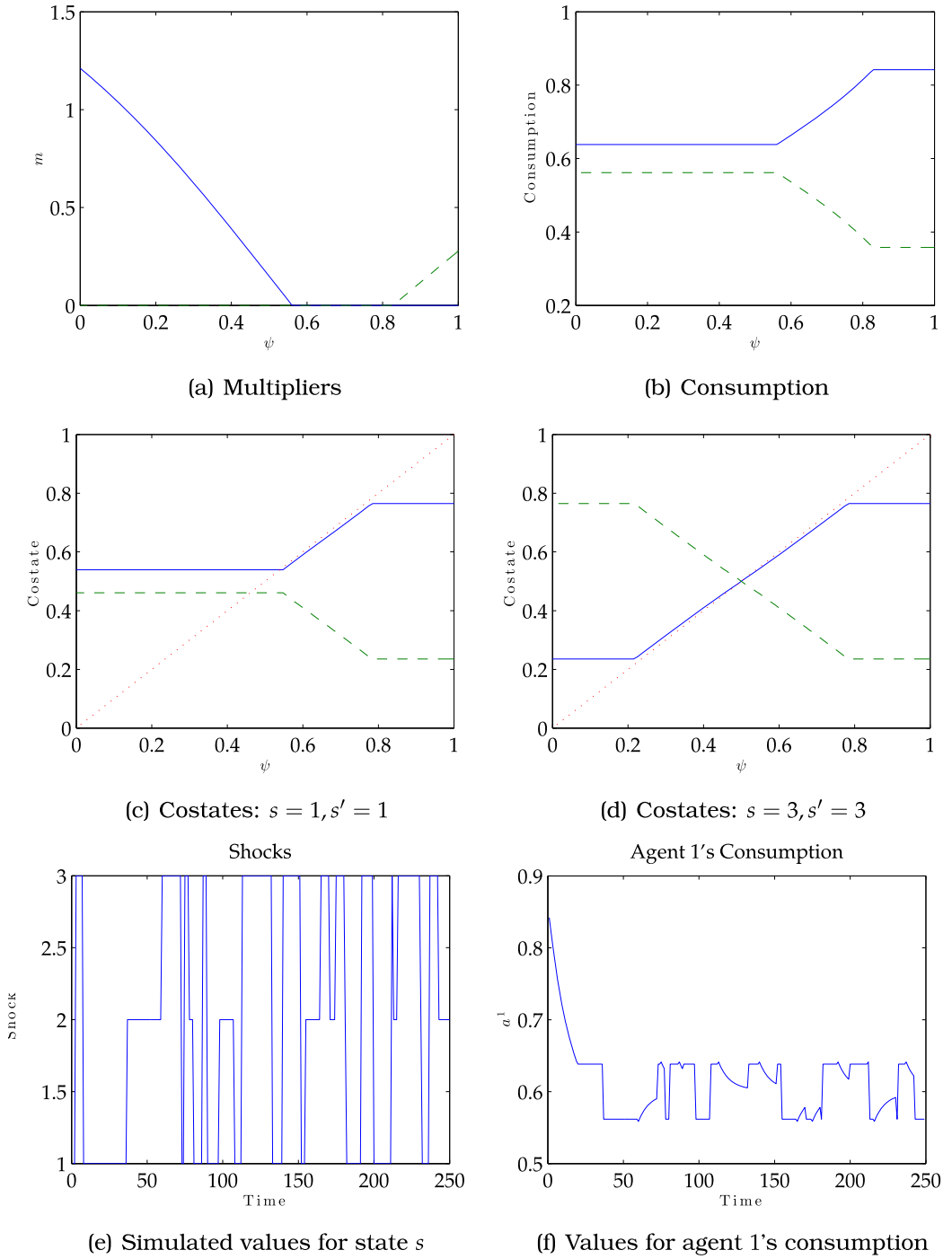


FIGURE 1.—In panels a–d, solid lines give agent 1's policy, dashed lines give agent 2's policy, and the 45 degree line is indicated by dots. Policies are given as functions of agent 1's normalized costate. Panel a shows multipliers on the no default constraints if $s = 1$; panel b, agent consumption in $s = 1$; panels c and d, the costates associated with remaining in state 1 and state 3, respectively. Panels e and f show a 250 period simulation. Shocks are given in the left panel, agent 1's consumption in the right.

Figure 1(a) gives the no default multipliers for both agents in state $s = 1$ as a function of the normalized costate for agent 1, $\psi = \frac{y^1}{y^1 + y^2}$. Since agent 1's outside option is high in state $s = 1$, her no default constraint binds and its multiplier is positive over a large range of low and middling values of ψ . In contrast, agent 2's multiplier is positive over a smaller range of high ψ values. Optimal consumption and continuation costate policy functions have a floor and ceiling structure: a floor is placed under an agent's consumption and continuation costate if its no default constraint binds; a ceiling is placed on these variables if the other agent's no default constraint binds. Figures 1(b) and (c) illustrate. The first shows agent consumption in $s = 1$, while the second shows continuation costates occurring if the economy remains in state $s = 1$. The floors and ceilings exhibited by these functions indicate the limits to consumption smoothing implied by the no default constraints: entering a period with a low costate and a high outside option, an agent's consumption and continuation costates must be jumped up to meet the floor and keep the agent from defaulting; entering a period with a high costate and a low outside option, an agent's consumption and continuation costates must be jumped down to satisfy the ceiling and keep the other agent from defaulting. The region of costate values between the floor and ceiling is interesting because it reveals additional dynamics that are present in the recursive, but not the expected utility case. Figure 1(c) shows that in this region, agent 1's future costate is slightly reduced, while agent 2's is slightly raised relative to their current levels. Hence, if shock $s = 1$ persists, agent 1's costate is gradually reduced until it hits the floor, while agent 2's is raised until it reaches the ceiling. This feature of the optimal risk sharing arrangement stems from the assumption that ρ is greater than σ , which implies that agents have a preference for the early resolution of risk and a dislike of cross-state variation in utility. The presence of a no default constraint for agent 2 impedes future utility smoothing across states and ensures that there is always a chance that, in the future, agent 1's utility decreases. Consequently, absent binding no default constraints on agent 1, it is profitable to tilt agent 1's utility profile towards the present and away from future shock state $s' = 1$ and, hence, reduce future period utility dispersion. Agent 1's normalized costate in future state $s' = 1$ is correspondingly reduced. This feature is absent under the standard assumption of expected utility¹⁷ and reversed if $\rho < \sigma$.

Figure 1(d) shows the response of costates to a repeated $s = 3$ shock. In the $s = 3$ state, both agents' outside options are low. As in Figure 1(c), binding no default constraints introduce floors and ceilings into the updated costate function (although now, the floor for agent 1 is at a lower level). Between this floor and ceiling, agent 1's updated costate function crosses the 45 degree line and has a slope less than 1. Again, absent binding no default constraints, recursive preferences introduce additional deterministic dynamics which, in this case, create a force for equality and equal division of the endowment.

Figures 1(e) and (f) report shocks and consumptions from a simulation of the optimal policy. Initially, agent 1's costate is such that, if shocks $s = 1$ and $s = 3$ are drawn, then neither agent's no default constraint binds. Consistent with the properties of the policy functions described above, draws of $s = 1$ cause agent 1's consumption to steadily fall towards the consumption floor for $s = 1$ (equal to 0.65) and remain there. After the draw of the first $s = 2$ shock, agent 2's no default constraint binds and agent 1's consumption is reduced to its consumption ceiling for this state (equal to 0.55). Thereafter, agent 1's

¹⁷See Alvarez and Jermann (2001) who found that policy functions for continuation utilities lie on the 45 degree line when no default constraints do not bind and agents have expected utility preferences.

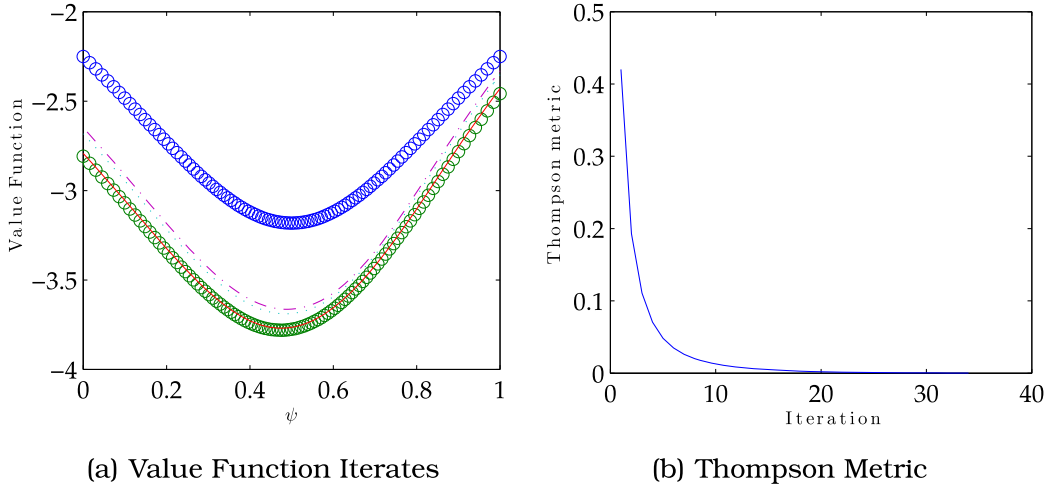


FIGURE 2.—Value function convergence. Panel a shows the calculated value function at iterations 1 (solid line), 10 (dotted line), and 25 (dash dot line) at $D(\psi, 1 - \psi)$. Bounding value functions $\underline{D}$ and $\overline{D}$ are shown with circles. Panel b reports Thompson metric distances between iterates.

consumption jumps on entering state $s = 1$ or 2 , while drifting towards 0.6 (and an equal sharing of the aggregate endowment) in $s = 3$.

Figure 2 shows successive value functions calculated during the value function iteration and the (Thompson metric) distance between them. The latter indicates convergence of the iterates.

2.4. Utility Promise Recursive Approaches

We conclude this section by describing an alternative utility promise-based recursive approach to solving the limited commitment problem. Define $\tilde{\mathcal{I}} := \{2, 3, \dots, n_I\}$ to exclude agent 1's label and let $\tilde{\mathcal{V}} := \prod_{i \in \tilde{\mathcal{I}}} [\underline{a}^i, \bar{a}^i]$ be a set of possible promises for agents in $\tilde{\mathcal{I}}$. For each $s \in \mathcal{S}$, let $\tilde{\mathcal{X}}^*(s)$ be the set of feasible initial payoffs for these agents (i.e., the set of payoffs for agents in $\tilde{\mathcal{I}}$ that are initial elements of a feasible promise plan given initial shock s).

Let $\tilde{P}^*(s_0, \tilde{v}_0)$ be the best possible feasible payoff to agent 1 given initial shock s_0 and feasible utility promise $\tilde{v}_0$ to the other agents. Problem (PP) can then be restated as¹⁸

$$P_0^* = \sup_{\tilde{v}_0 \in \tilde{\mathcal{X}}^*(s_0)} \lambda^1 \tilde{P}^*(s_0, \tilde{v}_0) + \sum_{\tilde{i}} \lambda^{\tilde{i}} \tilde{v}_0^{\tilde{i}}. \quad (9)$$

¹⁸This approach is suggested by a formulation of Kocherlakota (1996), who considered the case with two agents ($n_I = 2$), i.i.d. shocks, and expected utility preferences. See also Chapter 20 (Sections 20.7–20.10) in Ljungqvist and Sargent (2012) and the analysis of Alvarez and Jermann (2001). For a related approach, see Judd, Yeltekin, and Conklin (2003) and Sleet and Yeltekin (2016).

An optimal action and promise plan can be obtained by repeatedly solving

$$\begin{aligned}
 (a_t^*(s'), \tilde{v}_{t+1}^*(s', \cdot)) &= \underset{(a, \tilde{v}')}{\operatorname{argmax}} \left[(1 - \delta)[a^1]^{1-\sigma} + \delta \left\{ \sum_{s' \in \mathcal{S}} [\tilde{P}^*(\tilde{v}(s'))]^{1-\rho} \pi(s'|s_t) \right\}^{\frac{1-\sigma}{1-\rho}} \right]^{\frac{1}{1-\sigma}} \\
 \text{s.t. } \gamma(s_t) - \sum_{i \in \mathcal{I}} a^i &\geq 0; \quad \forall s', \quad \tilde{v}'(s') \in \tilde{\mathcal{X}}^*(s'); \\
 \forall i \in \tilde{\mathcal{I}}, \quad \tilde{v}_t^{i*}(s^t) &= \left[(1 - \delta)[a^i]^{1-\sigma} + \delta \left\{ \sum_{s' \in \mathcal{S}} [\tilde{v}^{i*}(s')]^{1-\rho} \pi(s'|s_t) \right\}^{\frac{1-\sigma}{1-\rho}} \right]^{\frac{1}{1-\sigma}} \geq w^i(s_t); \\
 \text{and } \left[(1 - \delta)[a^1]^{1-\sigma} + \delta \left\{ \sum_{s' \in \mathcal{S}} [\tilde{P}^*(\tilde{v}(s'))]^{1-\rho} \pi(s'|s_t) \right\}^{\frac{1-\sigma}{1-\rho}} \right]^{\frac{1}{1-\sigma}} &\geq w^1(s_t),
 \end{aligned} \tag{10}$$

starting from $(s_0, \tilde{v}_0^*)$. However, solution of (9) and (10) requires prior recovery of $(\tilde{\mathcal{X}}^*, \tilde{P}^*)$. One approach is to formulate $(\tilde{\mathcal{X}}^*, \tilde{P}^*)$ as the fixed point of a monotone (but not contractive) operator on a space of candidate value functions P and state space correspondences $\mathcal{X}$. This approach suggests an algorithm for computing $(\tilde{\mathcal{X}}^*, \tilde{P}^*)$ that relies on repeated application of the operator to a bounding pair $(\tilde{\mathcal{X}}_0, \tilde{P}_0)$ with $\tilde{\mathcal{X}}_0 \supset \tilde{\mathcal{X}}^*$ and $\tilde{P}_0 \geq P^*$. The algorithm is practical if there are two agents ($n_I = 2$), since then $\tilde{\mathcal{X}}^*$ is simply a family of intervals indexed by the shock. Thus, the iteration occurs on a space of value functions and end points for the intervals. For problems with $n_I > 2$, the iteration involves higher dimensional optimizations and approximation, repeated calculation, and updating of the state spaces in $\tilde{\mathcal{X}}_n(s) \subset \mathbb{R}^{n_I-1}$, $s = 1, \dots, n_S$. In addition, the operators used to update value functions and domains in these iterations, while monotone, are not contractions. Consequently, error bounds and convergence criteria are not available.¹⁹

3. THE DECISION MAKER'S PROBLEM

This section describes an abstract recursive choice problem that can be specialized to give many problems considered in the literature, including the limited commitment problem of Section 2. Other examples include optimal monetary policy problems and dynamic insurance with private information. Later sections formally develop the dual recursive method in this general abstract context.

Shocks and action plans. The process for shocks is denoted and defined as before. Transition matrix π is assumed to be strictly positively valued: for all $s, s' \in \mathcal{S}$, $\pi(s'|s) > 0$.²⁰ A nonempty set $\mathcal{A} \subset \mathbb{R}^{n_A}$ contains actions potentially available to a decision-maker at each date. The notation for an *action plan* remains: $\mathbf{a} = \{a_t\}_{t=0}^\infty$, with $a_0 \in \mathcal{A}$ and, $\forall t \geq 1$, $a_t : \mathcal{S}^t \rightarrow \mathcal{A}$. Let $\mathbf{A} = \mathcal{A}^\infty$ denote the set of action plans and $\mathbf{a}|s^t$ the s^t -continuation of action plan $\mathbf{a}$.

States and constraints. Let $\mathcal{V} \subset \mathbb{R}^{n_V}$ be a bounded set of “forward-looking” states similar to the utility promises of the last section and let $\mathbf{v} = \{v_t\}_{t=0}^\infty$, with $v_0 \in \mathcal{V}$ and for $t \geq 1$,

¹⁹A recent promising approach was suggested by Judd et al. (2017) who, in a different setting, relaxed the incentive constraints with penalty functions and used adaptive splines to prevent penalties proliferating.

²⁰The assumption of a finite number of shocks is maintained. This restriction and that of a positive transition avoid technical complications and streamline our presentation. Neither is essential for our main results.

$v_t : S^t \rightarrow \mathcal{V}$, be a plan for such states. To accommodate (amongst other things) agent utility promises consistent with non-expected utility, we assume that the law of motion for forward-looking states is constructed from a pair of functions $W^v : \mathcal{S} \times \mathcal{A} \times \mathbb{R}^{n_v} \rightarrow \mathbb{R}^{n_v}$ and $M^v : \mathcal{S} \times \mathcal{V}^{n_s} \rightarrow \mathbb{R}^{n_v}$. Plans $\mathbf{a}$ and $\mathbf{v}$ satisfy the law of motion if, for all $t \geq 0$ and $s^t \in S^t$,

$$v_t(s^t) = W^v[s_t, a_t(s^t), M^v[s_t, v_{t+1}(s^t, \cdot)]] \quad (11)$$

Further assume that for each $(s, a) \in \mathcal{S} \times \mathcal{A}$,²¹

$$W^v[s, a, M^v[s, \cdot]] : \mathcal{V}^{n_s} \rightarrow \mathcal{V}. \quad (12)$$

As in Section 2, forward-looking state variables often describe the payoffs of agents facing dynamic incentive constraints (with the decision-maker a planner maximizing a weighted sum of agent payoffs). W^v then corresponds to a “time aggregator” that gives the current state as a function of current actions and a certainty equivalent of future states, while M^v is a “stochastic aggregator” that generates the certainty equivalent. For example, if the problem involves a single agent with time additive-expected utility preferences, then $W^v[s, a, \mu] = (1 - \delta)u(s, a) + \delta\mu$ and $M^v[s, v'] = \sum_{s' \in \mathcal{S}} v'(s')\pi(s'|s)$.

We enrich the environment of the previous section by allowing for “backward-looking” states that have an initial condition and that are determined prior to the realization of the current shock. This enables us to accommodate capital into our framework. Given a bounded set $\mathcal{K} \subset \mathbb{R}^{n_K}$, a plan for backward-looking states (with initial condition $\bar{k}$) is a sequence: $\mathbf{k} = \{k_t\}_{t=0}^\infty$, with $k_0 = \bar{k}$ and for $t \geq 1$, $k_t : S^t \rightarrow \mathcal{K}$. Let $W^k : \mathcal{K} \times \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}^{n_K}$ define a law of motion for backward-looking states. An action plan $\mathbf{a}$ and a plan for backward-looking states $\mathbf{k}$ (with initial condition $\bar{k}$) satisfy this law of motion if, for all $t \geq 0$ and $s^{t+1} \in S^{t+1}$,²²

$$k_{t+1}(s^{t+1}) = W^k[k_t(s^t), s_t, a_t(s^t)]. \quad (13)$$

The image of W^k , $W^k[\mathcal{K}, \mathcal{S}, \mathcal{A}]$, is assumed to be a bounded set.

Additional constraints are constructed from actions and state variables according to, for all $t \geq 0$ and $s^t \in S^t$,

$$H[k_t(s^t), s_t, a_t(s^t), \{v_{t+1}(s^t, s')\}_{s' \in \mathcal{S}}] \geq 0, \quad (14)$$

where $H : \mathcal{K} \times \mathcal{S} \times \mathcal{A} \times \mathcal{V}^{n_s} \rightarrow \mathbb{R}^{n_H}$ is bounded. In applications, these inequalities capture incentive, resource, and, perhaps, other constraints. Define $\mathcal{K}^*(s)$ to be the set of k such that $(s, \bar{k})$ is the initial condition for some triple of plans $(\mathbf{k}, \mathbf{a}, \mathbf{v})$ satisfying (11), (13), and (14).

ASSUMPTION 1: For all $s \in \mathcal{S}$, $\mathcal{K}^*(s) \neq \emptyset$.

²¹ Many models incorporate stronger conditions that ensure a unique plan $\mathbf{v}$ can be associated with every $\mathbf{a}$. Most commonly, W^v is assumed to be Lipschitz continuous with respect to its third argument and M^v is assumed to be positively homogeneous with respect to its second argument. If, in addition, W^v is bounded with respect to $a \in \mathcal{A}$, then (12) holds and any plan $\mathbf{a}$ defines a unique plan $\mathbf{v}$. Marinacci and Montrucchio (2010) obtained a related result by assuming concavity of aggregators.

²²The law of motion (13) constrains k_{t+1} to be s^t -measurable. In our development of the dual recursive approach, it will be convenient to have this measurability restriction enforced by the law of motion constraints, rather than built into the definition of $\mathbf{k}$.

REMARK 1: Some problems such as those in Section 2 may lack backward-looking states. In these cases, backward-looking states are dropped from H and no law of motion for backward-looking states is associated with the problem. Simplified versions of the arguments that follow go through.

Decision-Maker's objective. The decision-maker maximizes a function of the initial values of the forward-looking states: $F : S \times \mathcal{V} \rightarrow \mathbb{R}$. Typical examples include situations, as in Section 2, in which a planner maximizes a weighted sum of incentive constrained agent payoffs and $F(s, v_0) = \lambda \cdot v_0$, with λ a vector of Pareto weights.

Problem statement. Define a *primal plan* $\mathbf{p}$ to be an action plan $\mathbf{a}$ augmented with a pair of plans for backward- and forward-looking states $\mathbf{k} = \{k_t\}_{t=0}^\infty$ and $\mathbf{v} = \{v_t\}_{t=0}^\infty$. The set of primal plans is given by

$$\mathbf{P} = \left\{ \mathbf{p} = (\mathbf{a}, \mathbf{k}, \mathbf{v}) \left| \begin{array}{l} a_0 \in \mathcal{A}, \forall t \geq 1, a_t : S^t \rightarrow \mathcal{A}, \\ k_0 \in \mathcal{K}, \forall t \geq 1, k_t : S^t \rightarrow \mathcal{K}, \\ v_0 \in \mathcal{V}, \forall t \geq 1, v_t : S^t \rightarrow \mathcal{V} \end{array} \right. \right\}.$$

We consider decision-maker problems of (or that can be expressed in) the form

$$P_0^* := \sup F[s_0, v_0] \quad (\text{P})$$

subject to $\mathbf{p} \in \mathbf{P}$, $k_0 = \bar{k}$, and $\forall t, s^t$,

$$k_{t+1}(s^{t+1}) = W^k[k_t(s^t), s_t, a_t(s^t)], \quad (15)$$

$$v_t(s^t) = W^v[s_t, a_t(s^t), M^v[s_t, v_{t+1}(s^t, \cdot)]], \quad \text{and} \quad (16)$$

$$H[k_t(s^t), s_t, a_t(s^t), v_{t+1}(s^t, \cdot)] \geq 0. \quad (17)$$

The limited commitment problem from Section 2 can be accommodated in this framework (by removing the backward-looking states and specializing the definitions of F , W^v , and M^v).

REMARK 2: Classical problems in economic dynamics (e.g., those considered in [Stokey, Lucas, and Prescott \(1989\)](#)) are accommodated by the framework described above. In these, the decision-maker's payoff is the only “forward-looking state variable,” $F[s, v_0] = v_0$, and this state variable does not appear in the function H . Further, for all $(s, k) \in S \times \mathcal{K}$, there is an $(a, k') \in \mathcal{A} \times \mathcal{K}$ such that $k' = W^k[k, s, a]$ and $H[k, s, a] \geq 0$. In these cases, repeated substitution of the law of motion for forward-looking states into the objective eliminates these variables completely.²³ Further, all members of the exogenously given $S \times \mathcal{K}$ are feasible initial states. Standard dynamic programming arguments can then be applied.

Assumption 2 provides sufficient conditions for the existence of an optimal plan.

²³Most commonly, $W^v[s, a, M^v[s, v']] = (1 - \delta)u(s, a) + \delta \sum_{s'} v'(s') \pi(s'|s)$ with a boundedness condition on u or, more generally, a terminal condition $\lim_{t \rightarrow \infty} \delta^t E[v_t]$. Then, by iteration, $v_0 = \sum_{t=0}^\infty \delta^t E[u(s_t, a_t)]$, and the forward-looking state payoff variables $\{v_t\}$ may be substituted out of the problem. The problem reduces to maximizing $\sum_{t=0}^\infty \delta^t E[u(s_t, a_t)]$ subject to, for each t and s^{t+1} , $a_t(s^t) \in \mathcal{A}$, $k_{t+1}(s^{t+1}) \in \mathcal{K}$ and $H[k_t(s^t), s_t, a_t(s^t)] \geq 0$.

ASSUMPTION 2—Continuity and Compactness: For all $s \in \mathcal{S}$, $W^v[s, \cdot, \cdot]$, $M^v[s, \cdot]$, and $W^k[\cdot, s, \cdot]$ are continuous, and $H[\cdot, s, \cdot, \cdot]$ and $F[s, \cdot]$ are upper semicontinuous. In addition, $\mathcal{A}$, $\mathcal{K}$, and $\mathcal{V}$ are compact.

PROPOSITION 1: Let Assumptions 1 and 2 hold and let $\bar{k}$ belong to $\mathcal{K}^*(s_0)$ in (P). Then $P_0^* > -\infty$ and an optimal solution $\mathbf{p}^*$ to (P) exists.

PROOF: See Appendix A.

Q.E.D.

4. RECURSIVE DUAL

This section begins by defining a Lagrangian for (P). The Lagrangian involves the product of an infinite number of constraint values and multipliers. To compress notation, constraint values are collected into an object called a *constraint process* and multipliers into an object called a *dual plan*. Definitions of these follow. The recursive dual approach is then developed.

4.1. Lagrangians and Dual Problems

Notation for the Lagrangian. A constraint process evaluates constraint functions inclusive of laws of motion in (15), (16), (17) at a given primal plan. For each primal plan $\mathbf{p}$, define $z_0^k(\mathbf{p}) := \bar{k} - k_0$ and, for all $t \geq 1$ and $s^t \in \mathcal{S}^t$, define

$$z_t^k(\mathbf{p})(s^t) := W^k[k_{t-1}(s^{t-1}), s_{t-1}, a_{t-1}(s^{t-1})] - k_t(s^t).$$

Then $\mathbf{z}^k(\mathbf{p}) = \{z_t^k(\mathbf{p})\}_{t=0}^\infty$ gives the values of the backward-looking law of motion constraints (inclusive of the initial condition) at $\mathbf{p}$. Similarly, define, for all $t \geq 0$, $s^t \in \mathcal{S}^t$,

$$z_t^v(\mathbf{p})(s^t) := W^v[s_t, a_t(s^t), M^v[s_t, v_{t+1}(s^t, \cdot)]] - v_t(s^t)$$

and $z_t^h(\mathbf{p})(s^t) := H[k_t(s^t), s_t, a_t(s^t), v_{t+1}(s^t, \cdot)]$. Then $\mathbf{z}^v(\mathbf{p}) = \{z_t^v(\mathbf{p})\}_{t=0}^\infty$ and $\mathbf{z}^h(\mathbf{p}) = \{z_t^h(\mathbf{p})\}_{t=0}^\infty$ give the values of the forward-looking law of motion and H constraints at $\mathbf{p}$. These terms are collected into the *constraint process* $\mathbf{z}(\mathbf{p}) = \{\mathbf{z}^k(\mathbf{p}), \mathbf{z}^v(\mathbf{p}), \mathbf{z}^h(\mathbf{p})\}$. The boundedness assumptions placed on primitives and the countable number of constraints ensure that for all $\mathbf{p} \in \mathbf{P} \subset \ell^\infty$, $\mathbf{z}(\mathbf{p}) \in \ell^\infty$, where ℓ^∞ is the set of sup-norm bounded sequences: $\{\{x_n\}_{n=1}^\infty | x_n \in \mathbb{R}^{n_K+n_V+n_H}, \sup_{n \in \mathbb{N}} \|x_n\|_E < \infty\}$, with $\|\cdot\|_E$ the relevant Euclidean norm.

A dual plan contains summable multipliers for the various constraints facing the decision-maker. Let $\mathbf{y}^k = \{y_t^k\}_{t=0}^\infty$, with $y_t^k : \mathcal{S}^t \rightarrow \mathbb{R}^{n_K}$, denote multipliers (costates) for the backward-looking law of motion and $\mathbf{y}^v = \{y_t^v\}_{t=0}^\infty$, with $y_t^v : \mathcal{S}^t \rightarrow \mathbb{R}^{n_V}$, multipliers (costates) for the forward-looking law of motion. Let $\mathbf{m} = \{m_t\}_{t=0}^\infty$, with $m_t : \mathcal{S}^t \rightarrow \mathbb{R}_+^{n_H}$, denote multipliers for the H -constraints. Collect these various multipliers into a *dual plan* $\mathbf{q} = \{\mathbf{m}, \mathbf{y}^k, \mathbf{y}^v\}$ and define the set of dual plans

$$\mathbf{Q} = \left\{ \mathbf{q} \left| \sum_{t=0}^\infty \sum_{s^t} \delta^t \{ \|m_t(s^t)\|_E + \|y_t^k(s^t)\|_E + \|y_t^v(s^t)\|_E \} \pi^t(s^t | s_0) < \infty \right. \right\},$$

with $\delta \in (0, 1]$ a normalizing discount.²⁴ Define the Lagrangian

$$\mathcal{L}(\mathbf{p}, \mathbf{q}) = F[s_0, v_0] + \langle \mathbf{q}, \mathbf{z}(\mathbf{p}) \rangle,$$

where

$$\langle \mathbf{q}, \mathbf{z}(\mathbf{p}) \rangle = \sum_{t=0}^{\infty} \sum_{s^t} \delta^t \{ m_t(s^t) \cdot z_t^h(\mathbf{p})(s^t) + y_t^k(s^t) \cdot z_t^k(\mathbf{p})(s^t) + y_t^v(s^t) \cdot z_t^v(\mathbf{p})(s^t) \} \pi^t(s^t | s_0)$$

is a weighted sum of constraint process terms. Since for each $\mathbf{p} \in \mathbf{P}$, $\mathbf{z}(\mathbf{p}) \in \ell^\infty$, it follows that for each $\mathbf{q} \in \mathbf{Q}$, $\langle \mathbf{q}, \mathbf{z}(\mathbf{p}) \rangle \in \mathbb{R}$.

Primal and dual problems. The decision-maker's problem (P) may be re-expressed as a primal sup-inf problem:

$$P_0^* := \sup_{\mathbf{p} \in \mathbf{P}} \inf_{\mathbf{q} \in \mathbf{Q}} \mathcal{L}(\mathbf{p}, \mathbf{q}). \quad (\text{SI})$$

Its dual interchanges the inf and sup operations:

$$D_0^* := \inf_{\mathbf{q} \in \mathbf{Q}} \sup_{\mathbf{p} \in \mathbf{P}} \mathcal{L}(\mathbf{p}, \mathbf{q}). \quad (\text{IS})$$

Discussion of the relation between these problems is deferred until Section 6. Instead, in the remainder of this section, a recursive formulation of (IS) is pursued.

4.2. Recursive Dual

The recursive dual approach decomposes (IS) into subproblems linked by costates.

Notation for the recursive dual. Following the notation of Section 2, let $y = (y^k, y^v) \in \mathcal{Y} := \mathbb{R}^{n_K + n_V}$ denote a pair of costate variables on current laws of motion. Let $q = (m, y')$ be a *current dual choice* with $m \in \mathbb{R}_+^{n_H}$ a current H -constraint multiplier and $y' = (y^{k'}, y^{v'}) \in \mathcal{Y}^{n_S}$ a tuple of continuation costates for the next period's laws of motion. Denote the set of current dual choices by $\mathcal{Q} = \mathbb{R}_+^{n_H} \times \mathcal{Y}^{n_S}$. Let $p = (a, k, v')$ be a *current primal choice* with $a \in \mathcal{A}$ a current action, $k \in \mathcal{K}$ a *current* backward-looking state, and $v' \in \mathcal{V}^{n_S}$ a tuple of *continuation* forward-looking states, one for each future shock s' . Let $\mathcal{P} := \mathcal{A} \times \mathcal{K} \times \mathcal{V}^{n_S}$ denote the set of current primal choices.

Extracting the first three (period 0 constraint) terms from $\langle \mathbf{q}, \mathbf{z}(\mathbf{p}) \rangle$, the Lagrangian in (IS) may be expanded as

$$\begin{aligned} D_0^* = \inf_{\mathbf{q} \in \mathbf{Q}} \sup_{\mathbf{p} \in \mathbf{P}} & F[s_0, v_0] - y_0^v \cdot \{ v_0 - W^v[s_0, a_0, M^v[s_0, v_1(\cdot)]] \} + y_0^k \cdot (\bar{k} - k_0) \\ & + m_0 \cdot H[k_0, s_0, a_0, v_1(\cdot)] + \delta \sum_{s_1 \in \mathcal{S}} \langle \mathbf{q}, \mathbf{z}(\mathbf{p}) | s_1 \rangle \pi(s_1 | s_0), \end{aligned} \quad (18)$$

where $\langle \mathbf{q}, \mathbf{z}(\mathbf{p}) | s_1 \rangle$ contains the shadow value of the constraint terms from period 1 after the realization of s_1 . Central to subsequent analysis is the following *continuation dual*

²⁴In most economic problems, the aggregator over forward states incorporates discounting over time and a certainty equivalent operator over future states. For these problems, it is convenient to normalize multipliers by the discount and by the probability distribution. We do so.

problem, which fixes the initial costates y_0 and removes the term $F[s_0, v_0] - y_0^v \cdot v_0 + y^k \cdot \bar{k}$ from the objective in (18):

$$\begin{aligned} D^*(s_0, y_0) = & \inf_{\mathbf{Q}(y_0)} \sup_{\mathbf{P}(v_0)} -y_0^k \cdot k_0 + y_0^v \cdot W^v[s_0, a_0, M^v[s_0, v_1(\cdot)]] \\ & + m_0 \cdot H[k_0, s_0, a_0, v_1(\cdot)] + \delta \sum_{s_1 \in \mathcal{S}} \langle \mathbf{q}, \mathbf{z}(\mathbf{p}) | s_1 \rangle \pi(s_1 | s_0). \end{aligned} \quad (19)$$

In (19), $\mathbf{Q}(y_0)$ omits $y_0 = (y_0^k, y_0^v)$ from $\mathbf{Q}$ and $\mathbf{P}(v_0)$ omits v_0 from $\mathbf{P}$. We call D^* the *dual value function*. We now derive some useful properties of D^* . In (19), D^* is defined via an inf-sup operation. In general, “inf-sup” operations on arbitrary functions can yield values of $-\infty$ or $+\infty$. However, we show below that if Assumption 1 holds, then D^* is real-valued. We also show that D^* is sub-linear, that is, convex and positively homogeneous, in costates. Both results have implications for the dual (co)state space which are described later in the section.

PROPOSITION 2: *For all $(s, y) \in \mathcal{S} \times \mathcal{Y}$, $D^*(s, y) < \infty$. If, in addition, Assumption 1 holds, then $D^*(s, y) > -\infty$ and $D^* : \mathcal{S} \times \mathcal{Y} \rightarrow \mathbb{R}$. Furthermore, for each $s \in \mathcal{S}$, $D^*(s, \cdot)$ is sub-linear, that is, convex and positively homogeneous of degree 1.*

PROOF: See Appendix A.

Q.E.D.

Collecting terms in (19) involving the initial current primal choice $p_0 = (k_0, a_0, v_1)$ gives the *current dual payoff* J :²⁵

$$\begin{aligned} J(s_0, y_0; q_0, p_0) = & -y_0^k \cdot k_0 + y_0^v \cdot W^v[s_0, a_0, M^v[s_0, v_1(\cdot)]] + m_0 \cdot H[k_0, s_0, a_0, v_1(\cdot)] \\ & - \delta \sum_{s_1 \in \mathcal{S}} y_1^v(s_1) \cdot v_1(s_1) \pi(s_1 | s_0) \\ & + \delta \sum_{s_1 \in \mathcal{S}} y_1^k(s_1) \pi(s_1 | s_0) \cdot W^k[k_0, s_0, a_0], \end{aligned} \quad (20)$$

where the terms in the second line of (20) are extracted from $\delta \sum_{s_1 \in \mathcal{S}} \langle \mathbf{q}, \mathbf{z}(\mathbf{p}) | s_1 \rangle \pi(s_1 | s_0)$ in (19). Proposition 3 relates D_0^* , D^* , and J and gives the key dynamic programming result for dual value functions.

PROPOSITION 3—Value Functions: *The value D_0^* satisfies*

$$D_0^* = \inf_{y \in \mathcal{Y}} \sup_{v \in \mathcal{V}} F[s_0, v] - y^v \cdot v + y^k \cdot \bar{k} + D^*(s_0, y^k, y^v). \quad (21)$$

Moreover, D^ satisfies the functional equation*

$$D^*(s, y) = \inf_{q \in \mathcal{Q}} \sup_{p \in \mathcal{P}} J(s, y; q, p) + \delta \sum_{s' \in \mathcal{S}} D^*(s', y'(s')) \pi(s' | s), \quad (22)$$

²⁵In economic applications, this function has an interpretation in terms of the shadow value of relaxing current constraints and current, past, and future laws of motion.

where $y'(s') = (y^{k'}, y^{v'})(s')$ and

$$\begin{aligned} J(s, y; q, p) = & -y^k \cdot k + y^v \cdot W^v[s, a, M^v[s, v']] + m \cdot H[k, s, a, v'] \\ & - \delta \sum_{s' \in \mathcal{S}} y^{v'}(s') \cdot v'(s') \pi(s'|s) \\ & + \delta \sum_{s' \in \mathcal{S}} y^{k'}(s') \pi(s'|s) \cdot W^k[k, s, a]. \end{aligned} \quad (23)$$

PROOF: See Appendix A.

Q.E.D.

The first stage problem (21) generates the initial costates; (22) then gives the dual Bellman equation. Moving from the dual problem (IS) to the recursive dual problems (21) and (22) involves interchanging an infimum operation over future dual variables with a supremum operation over current primal variables. The additive separability of the Lagrangian in these variables ensures that this interchange of operations does not alter the optimal value. See the proof of Proposition 3 for details.

REMARK 3: Our recursive dual formulation relies entirely on dual costate variables y , to summarize the past. Note that this implies that the supremum component of the inf-sup operations in (22) is static, that is, is of the form $J^*(s, y; q) := \sup_p J(s, y; q, p)$, and is embedded into a rather standard looking Bellman equation:

$$D^*(s, y) = \inf_{q \in \mathcal{Q}} J^*(s, y; q) + \delta \sum_{s' \in \mathcal{S}} D^*(s', y'(s')) \pi(s'|s). \quad (24)$$

REMARK 4: The primal “state” variables k and v continue to appear in the recursive dual problem. However, these variables are no longer passed between subproblems and in this sense no longer function as state variables. They are used to “penalize” choices of continuation costates and, hence, align them with current costates and multipliers.

Notice that, in the recursive dual framework, k and v are restricted to the exogenous set $\mathcal{K} \times \mathcal{V}^n$ and not the endogenous $\mathcal{X}^*$.

DEFINITION 1: Let $\mathcal{F}$ denote the set of functions $D : \mathcal{S} \times \mathcal{Y} \rightarrow \mathbb{R} \cup \{-\infty\}$. Define the dual Bellman operator $\mathcal{B} : \mathcal{F} \rightarrow \mathcal{F}$, $\forall (s, y) \in \mathcal{S} \times \mathcal{Y}$,

$$\mathcal{B}(D)(s, y) = \inf_{q \in \mathcal{Q}} \sup_{p \in \mathcal{P}} J(s, y; q, p) + \delta \sum_{s' \in \mathcal{S}} D(s', y'(s')) \pi(s'|s).$$

Theorem 1 recasts D^* as a fixed point of $\mathcal{B}$. It is a corollary of Proposition 3.

THEOREM 1: $D^* = \mathcal{B}(D^*)$.

The following lemma is a straightforward consequence of the definition of $\mathcal{B}$.

LEMMA 1—Monotonicity: If $D_1, D_2 \in \mathcal{F}$ and for all $(s, y) \in \mathcal{S} \times \mathcal{Y}$, $D_1(s, y) \geq D_2(s, y)$, then for all $(s, y) \in \mathcal{S} \times \mathcal{Y}$, $\mathcal{B}(D_1)(s, y) \geq \mathcal{B}(D_2)(s, y)$.

State spaces. Theorem 1 provides a dynamic programming formulation of the dual of the decision-maker's problem. It locates this dynamic programming problem on the state space of shocks and costates $\mathcal{S} \times \mathcal{Y}$ ($= \mathcal{S} \times \mathbb{R}^{n_K + n_Y}$). Proposition 2 showed that in the dual setting (with bounded primal variables and a nonempty feasibility set), the dual value function D^* is finite-valued on this set. Thus, the effective dual state space on which the value function is finite is immediately determined. In addition, $D^*(s, \cdot)$ is positively homogeneous of degree 1. This has the advantage that once the dual value functions $D^*(s, \cdot)$ are determined on the unit sphere $\mathcal{C} = \{y \in \mathcal{Y} \mid \|y\| = 1\}$, then they are determined everywhere via positive scaling. From a practical point of view, the state space may be identified with $\mathcal{S} \times \mathcal{C}$. In some problems, further refinement of the state space is natural and convenient. For example, in Section 2 and in many contracting and planner problems, it is natural to consider only nonnegative initial values for Pareto weights. It is then usual for costates to remain nonnegative and for the state space to be correspondingly restricted. Thus, in Section 2, the state space was set equal to $\mathcal{S} \times \mathcal{C}_+$ with $\mathcal{C}_+ = \{y \in \mathbb{R}_+^{n_K + n_Y} \mid \|y\|_E = 1\}$. Approximation of value functions on $\mathcal{C}$ (or $\mathcal{C}_+$) is discussed in Supplemental Material Appendix A. However, the homogeneity of candidate value functions combined with the unboundedness of the current dual set $\mathcal{Q}$ disrupts the conventional approach to proving that $\mathcal{B}$ is a contraction. This is addressed in Section 5.

Calculations. The dual Bellman operator involves an outer minimization over dual variables $q \in \mathcal{Q} = \mathbb{R}_+^{n_H} \times \mathcal{Y}^{n_S}$ and an inner maximization over primal variables $p \in \mathcal{P} = \mathcal{A} \times \mathcal{K} \times \mathcal{V}^{n_S}$ parameterized by q . These operations involve only simple constraints defined by product sets (i.e., $p \in \mathcal{P}$ and $q \in \mathcal{Q}$). Moreover, any additive separability in the functions W^k , W^v , and H can be exploited to decompose the inner maximization into component maximizations that can be run in parallel. In some cases, some or all of these component maximizations are sufficiently simple that they can be solved analytically. Thus, less and sometimes no inner step numerical maximization is needed. Additive separability may be across different pairs of actions and forward-looking states ($a^i, v'^i(\cdot)$), $i \in \mathcal{I}$ (e.g., corresponding to the actions and continuation payoffs of different agents); it may be across actions a and forward-looking states $v'(\cdot)$ (corresponding to time additive separability in payoffs) or across the components of $v'(s')$, $s' \in \mathcal{S}$ (corresponding to state additive separability in payoffs, that is, expected utility). There is also often additive separability between the backward-looking state and other actions. The examples in Section 2 and the quasilinear setup discussed below in Section 4.3 have different degrees of separability.

Policies. For arbitrary sets C and E and function $r : E \times C \rightarrow \mathbb{R} \cup \{-\infty, \infty\}$, define the argminmax operation:

$$\operatorname{argminmax}_{C|E} r(e, c) = \left\{ (c^*, e^*) \mid c^* \in \operatorname{argmin}_{c \in C} \sup_{e \in E} r(e, c) \text{ and } e^* \in \operatorname{argmax}_{e \in E} r(e, c^*) \right\}.$$

The solution to the dual problem (IS) is given by

$$\mathbf{A} := \operatorname{argminmax}_{\mathbf{Q}|\mathbf{P}} \mathcal{L}(\mathbf{p}, \mathbf{q}).$$

On the other hand, the solution to the recursive dual is described by a policy set:

$$G_0 = \operatorname{argminmax}_{\mathcal{Y}|\mathcal{V}} F[s_0, v] - y^v \cdot v + y^k \cdot \bar{k} + D^*(s_0, y^k, y^v)$$

and a policy correspondence

$$G(s, y) = \underset{\mathcal{Q}|\mathcal{P}}{\operatorname{argminmax}} J(s, y; m, y', p) + \delta \sum_{s' \in \mathcal{S}} D^*(s', y'(s')) \pi(s'|s).$$

The next proposition relates policies from the dual and the recursive dual.

PROPOSITION 4—Policies: $(\mathbf{q}^*, \mathbf{p}^*) \in \mathbf{A}$ only if $(y_0^*, v_0^*) \in G_0$ and for each $t \geq 1$, $s^t \in \mathcal{S}^t$, $(m_t^*(s^t), y_{t+1}^*(s^t), p_t^*(s^t)) \in G(s_t, y_t^*(s^t))$. Conversely, $(\mathbf{q}^*, \mathbf{p}^*) \in \mathbf{A}$ if $(y_0^*, v_0^*) \in G_0$, for each $t \geq 1$, $s^t \in \mathcal{S}^t$, $(m_t^*(s^t), y_{t+1}^*(s^t), p_t^*(s^t)) \in G(s_t, y_t^*(s^t))$ and

$$\liminf_{T \rightarrow \infty} \delta^{T+1} \sum_{s^{T+1}} D^*(s_{T+1}, y_{T+1}^*(s^{T+1})) \pi^{T+1}(s^{T+1}|s_0) \geq 0. \quad (\text{T})$$

PROOF: Appendix A.

Q.E.D.

4.3. The Quasilinear Case

Many problems have constraint functions that are quasilinear in k and v , that is,

$$W^k[k, s, a] = B^k(s)k - c(s, a),$$

$$W^v[s, a, M^v[s, v']] = (1 - \delta)u(s, a) + \delta \sum_{s' \in \mathcal{S}} B^v(s, s')v'(s')\pi(s'|s),$$

and

$$H[k, s, a, v'] = N^k(s)k + h(s, a) + \delta \sum_{s' \in \mathcal{S}} N^v(s, s')v'(s')\pi(s'|s),$$

for $\delta \in [0, 1)$, matrices $B^k(s)$, $N^k(s)$, $B^v(s, s')$, and $N^v(s, s')$, and functions $c : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}^{n_k}$, $u : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}^{n_u}$, and $h : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}^{n_h}$. Examples include the limited commitment problem discussed in Section 2 specialized to the case of expected utility preferences (i.e., $\rho = \sigma$) and all of the problems considered in Messner, Pavoni, and Sleet (2012) (each of which abstracts from backward-looking state variables). A Lagrangian of the form $\mathcal{L}$ (inclusive of laws of motion for states) may be associated with such problems and their recursive duals derived as in Proposition 3. However, directly exploiting quasilinear structure prior to the formulation of the Lagrangian leads to simplification. In particular, when the matrices B^r , $r \in \{k, v\}$, satisfy appropriate bounding conditions, the primal states k_t and v_t can be substituted out of the problem using their laws of motion. This leads to a modified Lagrangian and a simplified dual Bellman operator. In models without backward-looking primal states this operator is

$$\mathcal{B}(D)(s, y) = \inf_{m \in \mathbb{R}_+^H} \sup_{a \in \mathcal{A}} J(s, y; m, a) + \delta \sum_{s' \in \mathcal{S}} D(s', \phi(s, y, m, s')) \pi(s'|s),$$

with $\phi(s, y, m, s') = y \cdot B^v(s, s') + m \cdot N^v(s, s')$ and $J(s, y; m, a) = y \cdot u(s, a) + m \cdot h(s, a)$. Note that here continuation costates (for forward-looking primal states) are determined as a function of past costates and multipliers on current states. The dual value function D^* is a fixed point of this (modified) dual Bellman (by arguments essentially identical to those in Proposition 3).

5. CONTRACTION

This section establishes sufficient conditions for $\mathcal{B}$ to be contractive on an appropriate space of functions. The combination of an unbounded dual value function and an unbounded dual constraint set is an obstacle to conventional approaches to proving contractivity that rely on Blackwell's theorem.²⁶ Following [Marinacci and Montrucchio \(2010\)](#) and especially [Rinçon-Zapatero and Rodríguez-Palmero \(2003\)](#), we pursue a different approach. The idea is to use the monotonicity and concavity of $\mathcal{B}$ to show contractivity with respect to a Thompson-like metric, rather than a conventional sup-norm metric. Two positive valued functions g_1 and g_2 are said to be mutually scalable if there is a scaling factor $b \in \mathbb{R}_{++}$ satisfying $bg_1 \geq g_2$ and $bg_2 \geq g_1$. The Thompson-like metric evaluated at such a pair is the log of their smallest scaling factor. A version of this metric can be defined for an interval of possibly non-positive valued functions after a renormalization using the interval's bounding functions.

Towards showing the contractivity of $\mathcal{B}$, we first show that $\mathcal{B}$ preserves sub-linearity. For convenience, we will define $D : \mathcal{S} \times \mathcal{Y} \rightarrow \mathbb{R} \cup \{-\infty\}$ to be sub-linear if, for all $s \in \mathcal{S}$, $D(s, \cdot)$ is sub-linear.

LEMMA 2: *If $D : \mathcal{S} \times \mathcal{Y} \rightarrow \mathbb{R} \cup \{-\infty\}$ is sub-linear, then $\mathcal{B}(D)$ is sub-linear.*

PROOF: See Appendix B.

Q.E.D.

The key assumption ensuring contractivity is the following. As before, let $\mathcal{C} = \{y \in \mathcal{Y} \mid \|y\|_E = 1\}$ denote the unit sphere in $\mathbb{R}^{n_K + n_V}$.

ASSUMPTION 3: There is a triple of bounding functions $\underline{D} : \mathcal{S} \times \mathcal{Y} \rightarrow \mathbb{R}$, $\underline{D} : \mathcal{S} \times \mathcal{Y} \rightarrow \mathbb{R}$, and $\overline{D} : \mathcal{S} \times \mathcal{Y} \rightarrow \mathbb{R}$ and a number $\varepsilon > 0$ such that for all s , $\underline{D}(s, \cdot)$ is continuous and positively homogeneous of degree 1, $\overline{D}(s, \cdot)$ is continuous, and for all $(s, y) \in \mathcal{S} \times \mathcal{C}$, (i) $\underline{D}(s, y) + \varepsilon \leq \underline{D}(s, y) \leq \overline{D}(s, y)$, (ii) $\underline{D}(s, y) \leq \mathcal{B}(\underline{D})(s, y)$ and $\mathcal{B}(\overline{D})(s, y) \leq \overline{D}(s, y)$, and (iii) $\underline{D}(s, y) + \varepsilon < \mathcal{B}(\underline{D})(s, y)$.

We discuss the selection of bounding functions in the context of the limited commitment example in Supplemental Material Appendix B. However, note that if $\underline{D}$ satisfies Assumption 3(iii) and $\underline{D} \leq D^* \leq \overline{D}$, then, from the monotonicity of $\mathcal{B}$ and Theorem 1, for all $(s, y) \in \mathcal{S} \times \mathcal{C}$,

$$\underline{D}(s, y) + \varepsilon < \mathcal{B}(\underline{D})(s, y) \leq \mathcal{B}(\mathcal{B}(\underline{D}))(s, y) \leq \mathcal{B}(\mathcal{B}(D^*))(s, y) = D^*(s, y) \leq \overline{D}(s, y),$$

and $\underline{D}$ may be set equal to $\mathcal{B}(\underline{D})$. Given a triple of functions $\underline{D}$, $\underline{D}$, and $\overline{D}$ satisfying Assumption 3, let

$$\mathcal{G} = \{D : \mathcal{S} \times \mathcal{Y} \rightarrow \mathbb{R} \mid D \text{ is sub-linear and } \underline{D} \leq D \leq \overline{D}\}.$$

²⁶When the optimal value function is unbounded and the constraint correspondence compact-valued, it is often possible to prove contractivity on a space of weight norm bounded functions. In the dual setting, this approach is disrupted by the unboundedness of the constraint correspondence (for multipliers and costates).

Define the “Thompson-like” metric $d : \mathcal{G} \times \mathcal{G} \rightarrow \mathbb{R}_+$ according to, for each $D_1, D_2 \in \mathcal{G}$,

$$\begin{aligned} d(D_1, D_2) &= \sup_{s \times c} \left| \ln \left(\frac{D_1(s, y) - \underline{D}(s, y)}{\underline{D}(s, y) - \underline{\underline{D}}(s, y)} \right) - \ln \left(\frac{D_2(s, y) - \underline{D}(s, y)}{\underline{D}(s, y) - \underline{\underline{D}}(s, y)} \right) \right| \\ &\leq \sup_{s \times c} \ln \left(\frac{\overline{D}(s, y) - \underline{D}(s, y)}{\underline{D}(s, y) - \underline{\underline{D}}(s, y)} \right) \leq \sup_{s \times c} \ln \left(\frac{\overline{D}(s, y) - \underline{D}(s, y)}{\varepsilon} \right) < \infty, \end{aligned}$$

where the penultimate inequality and finiteness stem from Assumption 3.²⁷ That $(\mathcal{G}, d)$ is a complete metric space is shown next.

LEMMA 3: $(\mathcal{G}, d)$ is a complete metric space.

PROOF: See Appendix B.

Q.E.D.

Proposition 5 verifies that $\mathcal{B}$ is a contraction on $\mathcal{G}$. It relies on the concavity (and monotonicity) of $\mathcal{B}$ rather than any discounting-type conditions. This makes it well suited to the present setting where concavity of $\mathcal{B}$ is easy to show, but discounting (with respect to a suitable bounding norm) is not.

PROPOSITION 5: Let Assumption 3 hold. There is a $\rho \in [0, 1)$ such that, for all $D_1, D_2 \in \mathcal{G}$, $d(\mathcal{B}(D_1), \mathcal{B}(D_2)) \leq \rho d(D_1, D_2)$, that is, $\mathcal{B}$ is a contraction on $(\mathcal{G}, d)$ with modulus of contraction ρ .

PROOF: See Appendix B.

Q.E.D.

Application of the contraction mapping theorem implies that $\mathcal{B}$ has a unique fixed point in $\mathcal{G}$.

THEOREM 2: Let Assumption 3 hold and assume that $\underline{D} \leq D^* \leq \overline{D}$. D^* is the unique fixed point of $\mathcal{B}$ in $\mathcal{G}$. Also, there is a $\rho \in [0, 1)$ such that, for any $D_0 \in \mathcal{G}$, $\mathcal{B}^n(D_0) \xrightarrow{d} D$ with $d(\mathcal{B}^n(D_0), D^*) \leq \rho^n d(D_0, D^*) \leq \rho^n d(\overline{D}, \underline{D})$.

PROOF: From Lemma 2, D^* is sub-linear and by assumption it is bounded below by $\underline{D}$ and above by $\overline{D}$. Thus, $D^* \in \mathcal{G} \neq \emptyset$. Also by Lemma 2, if $D \in \mathcal{G}$, then $\mathcal{B}(D)$ is sub-linear and by the monotonicity of $\mathcal{B}$ and Assumption 3 it is bounded below by $\underline{D}$ and above by $\overline{D}$. Thus, $\mathcal{B} : \mathcal{G} \rightarrow \mathcal{G}$. By Proposition 5, it is contractive on $\mathcal{G}$. The desired results then stem from the contraction mapping theorem.

Q.E.D.

REMARK 5: An immediate consequence of the previous (uniqueness) result is the following “Verification Theorem”: Under the assumptions of Theorem 2, if $\hat{D}$ is a fixed point of $\mathcal{B}$ in the space of sub-linear functions bounded below by $\underline{D}$ and above by $\overline{D}$, then $\hat{D} = D^*$.

REMARK 6: For application of Theorem 2, it is sufficient to know (i) that bounding functions satisfying Assumption 3 and $\underline{D} \leq D^* \leq \overline{D}$ exist and (ii) that a given sublinear

²⁷ In particular, $\underline{D}(s, y) - \underline{\underline{D}}(s, y) \geq \varepsilon > 0$ for all (s, y) .

function D_0 lies between $\underline{D}$ and $\overline{D}$ and can thus serve as an initial condition in a value iteration. Explicit calculation of the bounding functions is unnecessary. This contrasts with results relying on monotone (not contractive) operators, which require an upper or lower bound to the true value function as an initial condition. In addition, as always, the contraction result allows us to calculate error bounds and rates of convergence and is, thus, an improvement on results relying only on monotone iterations and pointwise convergence of iterates.

6. RELATING ORIGINAL AND DUAL PROBLEMS

The preceding sections established that the recursive dual supplies the optimal value and solutions for the dual problem (IS). Consequently, *if* the dual problem supplies the optimal value and an optimal plan for the original primal problem (P), then the recursive dual does as well and the original problem may be solved via dual value iteration. This section discusses conditions for the dual and the original problems to have common optimal values and policies. It begins by recalling the classical weak duality inequality, which requires no additional assumptions. It then gives conditions in terms of the existence and properties of saddle points of the Lagrangian $\mathcal{L}$. Next, it gives sufficient conditions for saddle existence (and, hence, equality of optimal values and necessity of primal solutions for dual problems) in terms of primitives. Finally, a numerical procedure is described for checking whether a primal plan obtained via the recursive dual approach solves the original primal problem.

6.1. Saddles and Recursive Dual Policies

Without further restriction, classical weak duality implies that the optimal dual value bounds the optimal primal value: $D_0^* \geq P_0^*$. Thus, with no further assumptions, the recursive dual gives welfare bounds for optimal policies or policy improvements.

A well-known sufficient condition for equality of optimal values, albeit not on primitives, is that the Lagrangian admits a saddle point, that is, there is a pair $(\mathbf{p}^*, \mathbf{q}^*) \in \mathbf{P} \times \mathbf{Q}$ such that for all $\mathbf{p} \in \mathbf{P}$ and $\mathbf{q} \in \mathbf{Q}$, $\mathcal{L}(\mathbf{p}, \mathbf{q}^*) \leq \mathcal{L}(\mathbf{p}^*, \mathbf{q}^*) \leq \mathcal{L}(\mathbf{p}^*, \mathbf{q})$. Saddle existence also ensures that if $\mathbf{p}^*$ solves (P) and $\mathbf{q}^*$ solves the dual (i.e., attains the minimum in (IS)), then $(\mathbf{q}^*, \mathbf{p}^*) \in \Lambda$. The following proposition summarizes the situation.

DEFINITION 2: A primal plan $\hat{\mathbf{p}}$ is consistent with the recursive dual policy if there is a $\hat{\mathbf{q}}$ such that $(\hat{y}_0, \hat{v}_0) \in G_0$ and for $t \geq 1$, $s^t \in \mathcal{S}^t$, $(\hat{m}_t(s^t), \hat{y}_{t+1}(s^t), \hat{p}_t(s^t)) \in G(s_t, \hat{y}_t(s^t))$.

PROPOSITION 6: *Assume that $\mathcal{L}$ admits a saddle point. Then the optimal dual and primal values are equal: $D_0^* = P_0^*$. In addition, if $\mathbf{p}^*$ solves (P), then it is consistent with the recursive dual policy.*

PROOF: See Appendix C.

Q.E.D.

Proposition 6 only requires that $\mathcal{L}$ admits a saddle point. It does *not* require that the Lagrangian associated with every $(s^t, y_t(s^t))$ -continuation problem has a saddle point, as is the case in [Marcet and Marimon \(2011\)](#). Proving, or numerically checking, the existence of a saddle point for $\mathcal{L}$, while nontrivial, is less demanding than doing so for all possible histories.

From Proposition 6, if a saddle exists, then the dual recursive problem gives the optimal primal value and optimal primal plans are consistent with the recursive dual policy.

However, other (suboptimal) plans may also be consistent with the recursive dual policy. Stronger conditions are needed to ensure that the “finite penalization” implicit in the dual problem is “sharp enough” to pin down only primal solutions.²⁸ A simple way to obtain a sufficiency result for policies is to require that the set of plans consistent with the recursive dual policy is unique.²⁹

PROPOSITION 7: *Assume that $\mathcal{L}$ has a saddle point. Let $\hat{\mathbf{P}}$ be the set of primal plans that are consistent with the recursive dual policy. If $\hat{\mathbf{p}}$ is the unique element of $\hat{\mathbf{P}}$, then $\hat{\mathbf{p}}$ is the unique solution of (P).*

PROOF: See Appendix C.

Q.E.D.

6.2. Concave Problems

For problems with *only* inequality constraints, a well-known sufficient condition for equality of optimal values and a minimizing dual multiplier (so-called “strong duality”) is that (i) the objective and constraint functions are defined on a convex domain and are concave and (ii) the evaluation of the constraints at some primal choice lies in the interior of the constraint space’s closed nonnegative cone (a Slater condition). If, in addition, a solution to the optimization exists, then it and the minimizing multiplier constitute a saddle point.

There are two difficulties in applying these results to our setting. First, (P) incorporates equality constraints describing the laws of motion for states. We deal with this below through a combination of relaxation and substitution. In many economic problems, monotonicity assumptions on nonlinear laws of motion for states are natural and permit relaxation of the corresponding constraints. We present one specific set of monotonicity assumptions and then discuss other possibilities. Further forward-looking state variables with quasilinear laws of motion can be directly substituted from the problem and their laws of motion removed as constraints. After relaxation and substitution, the classical sufficient conditions for strong duality given above are applicable. However, a second difficulty remains. It stems from the fact that these sufficient conditions ensure multiplier existence in the dual of the constraint space. In our setting, this is $\ell^{\infty,*}$, the set of all continuous linear functionals on ℓ^{∞} , not the more convenient space of summable sequences ℓ^1 which our dual formulation requires and on which $\mathcal{L}$ is defined.³⁰ However, as we show below, our constraint structure ensures that if a minimizing multiplier exists in $\ell^{\infty,*}$, then one exists in ℓ^1 as well.

We partition the elements of the forward-looking state v into two groups: $v = (v^c, v^l)$ and assume that $\mathcal{V} = \mathcal{V}^c \times \mathcal{V}^l$, with $v^r \in \mathcal{V}^r \subset \mathbb{R}^{n_r}$, $r = c, l$, and each $\mathcal{V}^r$ bounded. In what follows, v^c corresponds to a forward-looking state with a (fully) nonlinear law of motion and v^l corresponds to a forward-looking state with a quasilinear law of motion. As indicated above, these different types of forward-looking states are handled differently in the analysis below. Denote the aggregators describing laws of motion for them by W^r and M^r ,

²⁸This issue was emphasized by Messner and Pavoni (2015). Cole and Kubler (2012) described a procedure for augmenting a recursive saddle point problem with lotteries over the extreme points of flat regions of the continuation value function that permits recovery of an optimal primal solution. It remains to integrate their approach into our formulation.

²⁹Note that uniqueness of multipliers is not needed.

³⁰Strictly, $\mathcal{L}$ is defined on a set of discounted and weighted multipliers. However, the unnormalized multipliers are in ℓ^1 and $\mathcal{L}$ can be redefined to have $\ell^1 \times \mathbf{P}$ as its domain.

for $r \in \{c, l\}$. Corresponding to the earlier requirement that $W^v[s, a, M^v[s, \cdot]] : \mathcal{V}^{ns} \rightarrow \mathcal{V}$, assume $W^r[s, a, M^r[s, \cdot]] : (\mathcal{V}^r)^{ns} \rightarrow \mathcal{V}^r$, $r = c, l$. Finally, for $r = c, l$, let $v^r = \{v^r(s)\}_{s \in \mathcal{S}} \in (\mathcal{V}^r)^{ns}$. The following monotonicity, concavity, and quasilinearity assumptions are imposed upon the problem.

ASSUMPTION 4—Monotonicity: (i) $F[s_0, \cdot]$ is increasing in v^c . (ii) $\forall (s, a) \in \mathcal{S} \times \mathcal{A}$, $W^c[s, a, \cdot]$, $M^c[s, \cdot]$, and $W^k[\cdot, s, a]$ are increasing. In addition, if $W^k[k, s, a] \in \mathcal{K}$, then for all $k' \in \mathcal{K}$ with $k' > k$, $W^k[k', s, a] \in \mathcal{K}$. (iii) $\forall (s, a, v^l) \in \mathcal{S} \times \mathcal{A} \times (\mathcal{V}^l)^{ns}$, $H[\cdot, s, a, \cdot, v^l]$ is increasing in (k, v^c) .

ASSUMPTION 5—Concavity and Quasilinearity: (i) $F[s_0, \cdot]$ is concave in all arguments and linear in $v_0^l \in \mathbb{R}^{n_l}$. (ii) $\forall s \in \mathcal{S}$, $H[\cdot, s, \cdot]$ is concave in all arguments and linear in v^l . (iii) $\forall s \in \mathcal{S}$, $W^c[s, \cdot]$, $M^c[s, \cdot]$, and $W^k[s, \cdot]$ are concave in all arguments. (iv) $\forall (s, a, v^l)$, $W^l[s, a, M^l[s, v^l]] = (1 - \delta)u^l(s, a) + \delta \sum_{s' \in \mathcal{S}} B^l(s, s')v^l(s')\pi(s'|s)$, with $\delta \in [0, 1)$, $u^l : \mathcal{S} \times \mathcal{A} \rightarrow \mathcal{V}^l$ a concave function, $\mathcal{V}^l \subset \mathbb{R}^{n_l}$ a bounded set, and each $B^l(s, s')$ a diagonal matrix of size n_l with all elements less than or equal to 1 in absolute value.

We also impose a Slater condition. Let $\mathbf{z}^j(\mathbf{p}) = \{z_t^j(\mathbf{p})\}_{t=0}^\infty$, $j = k, h, l, c$, denote the constraint process terms, with for $j = l, c$, $z_t^j(\mathbf{p})(s^t) = -v_t^j(s^t) + W^j[s_t, a_t(s^t), M^j[s_t, v_{t+1}^j(s^t, \cdot)]]$.

ASSUMPTION 6—Slater: There is a $\hat{\mathbf{p}} \in \mathbf{P}$ such that $\mathbf{z}^l(\hat{\mathbf{p}}) = 0$ and $\inf_{t, s^t} z_t^j(\hat{\mathbf{p}})(s^t) > 0$ for $j = k, c, h$.

Proposition 8 establishes saddle point existence given a solution to the original problem (P) and Assumptions 4 to 6. Note that the multiplier component of the saddle point is in ℓ^1 rather than the larger and less convenient set $\ell^{\infty, *}$.

PROPOSITION 8: *Let Assumptions 2 and 4 to 6 hold and let $\bar{k} \in \mathcal{K}^*(s_0)$. Then $\mathcal{L}$ admits a saddle point. If $\mathbf{p}^*$ is the primal component of a saddle point and $\mathbf{q}^* \in \ell^1$ solves (IS₀), then $\mathcal{L}(\mathbf{p}^*, \mathbf{q}^*) = D_0^* = P_0^*$ and the elements of $(\mathbf{p}^*, \mathbf{q}^*)$ satisfy $(y_0^{k*}, y_0^{v*}, v_0^*) \in G_0$ and for all $t \geq 1$ and $s^t \in \mathcal{S}^t$, $(m_t^l(s^t), y_{t+1}^*(s^t), p_t^*(s^t)) \in G(s_t, y_t^*(s^t))$.*

PROOF: See Appendix C.

Q.E.D.

REMARK 7: Proposition 8 applies to problems with a mixture of nonlinear forward- and backward-looking states and quasilinear forward-looking states. Assumptions 4 and 5 may be reduced for simpler problems. For example, if $\rho \neq \sigma$, then the limited commitment problem in Section 2 involves only nonlinear forward-looking state variables and those parts of Assumptions 4 and 5 relating to backward-looking or quasilinear forward-looking constraints may be dropped. If $\rho = \sigma$, then preferences are specialized to the expected utility form and, hence, the limited commitment problem involves only quasilinear forward-looking constraints. Then Assumption 4 and parts of Assumption 5 are not needed.

The role of Assumption 4 is to permit a relaxation of the nonlinear law of motion constraints. Assumption 4(i) and (ii) are satisfied in all settings in which the forward-looking states are payoffs and the backward-looking states are capital (and output is increasing in capital). Hence, they are quite natural. Assumption 4(iii) is more restrictive. Specifically, this assumption requires that H is increasing in the backward-looking state k . While

appropriate for some settings, it need not hold in others. For example, it need not hold in limited commitment settings with capital accumulation. In these settings, additional capital increases both output *and* the outside option values of agents. Consequently, the no default constraints embedded in H may decrease over certain capital ranges. On the other hand, if agent utilities are increasing in consumption, then it is never desirable to discard output and select an allocation in which the law of motion for capital is relaxed, that is, an allocation in which $k_{t+1}(s^{t+1}) < W^k[k_t(s^t), s_t, a_t(s^t)] := f(k_t(s^t)) - \sum_{i \in \mathcal{I}} a_t^i(s_t)$, with f the production function. Instead, consumption will be raised, so relaxing both current and past no default constraints. Thus, in this setting, relaxation of the laws of motion for capital and (if agents have non-expected utility preferences) agent utility does not raise the optimal payoff or expand the solution set. This example motivates an alternative to Assumptions 4 and 5 in which Assumption 4(ii) is supplemented with the requirement that, for all (k, s) , $W^k[k, s, \cdot]$ is decreasing in a and $W^e[s, \cdot, m]$ is increasing in a , Assumption 4(iii) is modified to require that $H[k, s, \cdot, \cdot, \cdot]$ is increasing in a , v^c , and v^l , and Assumption 5 is strengthened to require that $u^l(s, \cdot)$ is increasing in a and the matrix B^l is positive. With these modifications, Proposition 8 is applicable to limited commitment problems with capital accumulation such as the one considered by Cooley, Marimon, and Quadrini (2004).

6.3. Ex Post Check

The following proposition describes a numerical procedure for checking whether a primal plan obtained via the recursive dual approach solves the original primal problem. The procedure does not require any concavity assumptions on primitives. We call a pair $(\hat{\mathbf{p}}, \hat{\mathbf{q}})$ a *candidate plan* if it is obtained from the recursive dual policy correspondence: $(\hat{q}_0^k, \hat{q}_0^v, \hat{v}_0) \in G_0$, and $\forall t \geq 1$ and $s^t \in S^t$, $(\hat{q}_t(s^t), \hat{p}_t(s^t)) \in G(s_t, \hat{y}_t(s^t))$.

PROPOSITION 9: *If $(\hat{\mathbf{p}}, \hat{\mathbf{q}})$ is a candidate plan satisfying (i) Condition (T) in Proposition 4, (ii) $-\hat{y}_0^v \cdot \hat{v}_0 + \hat{y}_0^k \cdot \bar{k} + D^*(s_0, \hat{y}_0^k, \hat{y}_0^v) \leq 0$, and (iii) feasibility of $\hat{\mathbf{p}}$ for (P), then $\hat{\mathbf{p}}$ is optimal for (P). In addition, $(\hat{\mathbf{p}}, \hat{\mathbf{q}})$ is a saddle point for $\mathcal{L}$.*

PROOF: See Appendix C.

Q.E.D.

Despite its simplicity, Proposition 9 is the basis of a useful ex post check of primal optimality. Suppose the recursive dual problem has been solved and a fixed point $\hat{D}$ of the operator $\mathcal{B}$ obtained. If Assumption 3 and the condition $\underline{D} \leq D^* \leq \overline{D}$ are satisfied and $\hat{D}$ lies between the bounding functions $\underline{D}$ and $\overline{D}$, then by Theorem 2, $\hat{D} = D^*$. If the solution of the recursive dual delivers a candidate plan $(\hat{\mathbf{p}}, \hat{\mathbf{q}})$, then Proposition 9 provides sufficient conditions for $\hat{\mathbf{p}}$ to be a solution to (P) and for the existence of a saddle point of the associated Lagrangian. In practice, the value function D^* and the policy correspondence G must be approximated via, say, a numerical implementation of the value iteration described in Theorem 2, and the conditions in Proposition 9 checked numerically to within some acceptable level of tolerance.

7. CONCLUSION

In many settings, the state space of a dynamic economic problem is defined implicitly and must be recovered as part of the solution to the problem. This complicates the application of recursive methods. Associated dual problems have recursive formulations in

which costates are used to keep track of histories of past or feasible future actions. The dual (co-)state space is often given a priori. Despite the unboundedness of the dual value functions and the lack of a bounded constraint correspondence, contractivity of the dual Bellman operator (with respect to Thompson-like metric) may be established if suitable bounding functions are available. In many problems, they are.

APPENDIX A: PROOFS FOR SECTION 3 AND SECTION 4

PROOF OF PROPOSITION 1: Endow $\mathbf{P}$ with the product topology. Given s_0 , for all $k \in \mathcal{K}$, let $\mathbf{C}(s_0, k) := \{\mathbf{p} \in \mathbf{P} | k_0 = k, v_0 \in \mathcal{V}, \text{ and } \forall t \geq 1, \dots, s^t \in \mathcal{S}^t(15), (16), \text{ and } (17) \text{ hold}\}$.

LEMMA 4: *Given Assumption 2, $\forall k \in \mathcal{K}$, $\mathbf{C}(s_0, k)$ is compact in the product topology.*

PROOF: Under Assumption 2, $\mathcal{A} \times \mathcal{K} \times \mathcal{V}^{n_s}$ is compact. Hence, by the Tychonoff theorem, $\mathcal{V} \times \mathcal{A} \times (\mathcal{A} \times \mathcal{K} \times \mathcal{V}^{n_s})^\infty$ is compact in the product topology. Since $\mathbf{C}(s_0, k) \subset \mathcal{V} \times \mathcal{A} \times (\mathcal{A} \times \mathcal{K} \times \mathcal{V}^{n_s})^\infty$ and closed subsets of compact spaces are compact, it suffices to show that $\mathbf{C}(s_0, k)$ is closed. Let $\{\mathbf{p}^n\}_{n=1}^\infty$ denote a sequence with each $\mathbf{p}^n \in \mathbf{C}(s_0, k)$ and such that $\mathbf{p}^n$ converges in the product topology to $\mathbf{p}$. Then, for all t and s^t , the corresponding components of $\mathbf{p}^n$ converge to those of $\mathbf{p}$. The compactness of $\mathcal{A}$, $\mathcal{K}$, and $\mathcal{V}$, continuity of W^k and W^v , and upper semicontinuity of H then ensure that $\mathbf{p}$ is in $\mathbf{P}$ and satisfies (15), (16), and (17). Thus, $\mathbf{p} \in \mathbf{C}(s_0, k)$. *Q.E.D.*

The planner's problem is: $P_0^* = \sup_{\mathbf{p} \in \mathbf{C}(s_0, \bar{k})} F(s_0, v_0)$. By assumption, $F(s_0, \cdot)$ is upper semicontinuous. Since $\bar{k} \in \mathcal{K}^*(s_0)$ and by Assumption 1, $\mathbf{C}(s_0, \bar{k}) \neq \emptyset$. By Lemma 4, $\mathbf{C}(s_0, \bar{k})$ is compact. The desired result follows from Weierstrass's theorem. *Q.E.D.*

PROOF OF PROPOSITION 2: To show $D^* < \infty$, choose an arbitrary $(s, y) \in \mathcal{S} \times \mathcal{Y}$. Then,

$$\begin{aligned} D^*(s, y) &= \inf_{\mathbf{Q}(y)} \sup_{\mathbf{P}(v_0)} -y^k \cdot k_0 + y^v \cdot W^v[s, a_0, M^v[s, v_1(\cdot)]] \\ &\quad + m_0 \cdot H[k_0, s, a_0, v_1] + \delta \sum_{s_1 \in \mathcal{S}} \langle \mathbf{q}, \mathbf{z}(\mathbf{p}) | s_1 \rangle \pi(s_1 | s) \\ &\leq \sup_{p \in \mathcal{P}} -y^k \cdot k_0 + y^v \cdot W^v[s, a_0, M^v[s, v_1(\cdot)]] < \infty, \end{aligned} \tag{25}$$

where the first inequality uses $0 \in \mathbf{Q}(y)$ (and, hence, its feasibility for the infimum) and the second inequality uses the boundedness of $\mathcal{K}$ and $\mathcal{V}$ and $W^v[s, \cdot, M^v[s, \cdot]] : \mathcal{A} \times \mathcal{V}^{n_s} \rightarrow \mathcal{V}$. Since (s, y) was arbitrary, $D^* < \infty$.

To show that $D^* > -\infty$, note that Assumption 1 guarantees that, for all s , there is a $\hat{k}_0 \in \mathcal{K}^*(s)$ and, so, a feasible primal plan $\hat{\mathbf{p}} \in \mathbf{P}$ starting at $(s, \hat{k}_0)$. Hence, for arbitrary $\mathbf{q} \in \mathbf{Q}(y)$,

$$\begin{aligned} &\sup_{\mathbf{P}} -y^k \cdot k_0 + y^v \cdot W^v[s, a_0, M^v[s, v_1(\cdot)]] + m \cdot H[k_0, s, a_0, v_1(\cdot)] \\ &\quad + \delta \sum_{s_1 \in \mathcal{S}} \langle \mathbf{q}, \mathbf{z}(\mathbf{p}) | s_1 \rangle \pi(s_1 | s) \\ &\geq -y^k \cdot \hat{k}_0 + y^v \cdot W^v[s, \hat{a}_0, M^v[s, \hat{v}_1(\cdot)]] > -\infty, \end{aligned}$$

where the first inequality uses the fact that $\hat{\mathbf{p}}$ is feasible, but not necessarily maximal at $\mathbf{q}$. Since $\mathbf{q}$ was arbitrary, $-y^k \cdot \hat{k}_0 + y^v \cdot W^v[s, \hat{a}_0, M^v[s, \hat{v}_1(\cdot)]]$ is a finite lower bound for $\sup_{\mathbf{p}} -y^k \cdot k_0 + y^v \cdot W^v[s, a_0, M^v[s, v_1(\cdot)]] + m \cdot H[k_0, s, a_0, v_1(\cdot)] + \delta \sum_{s_1 \in \mathcal{S}} \langle \mathbf{q}, \mathbf{z}(\mathbf{p}) | s_1 \rangle \times \pi(s_1 | s)$ and, so, for $D^*(s, y)$. Since (s, y) was arbitrary, $D^* > -\infty$.

For the proof of sub-linearity of D^* , see the proof of Lemma 2 below.

Q.E.D.

PROOF OF PROPOSITION 3: We have

$$\begin{aligned}
 D_0^* &= \inf_{\mathbf{Q}} \sup_{\mathbf{P}} \mathcal{L}(\mathbf{p}, \mathbf{q}) \\
 &= \inf_{\mathcal{Y}} \inf_{\mathbf{Q}(\mathcal{Y})} \sup_{\mathcal{V}} \sup_{\mathbf{P}(v_0)} F[s_0, v_0] + y_0^k \cdot (\bar{k} - k_0) \\
 &\quad + y_0^v \cdot (W^v[s_0, a_0, M^v[s_0, v_1(\cdot)]] - v_0) \\
 &\quad + m_0 \cdot H[k_0, s_0, a_0, v_1(\cdot)] + \delta \sum_{s_1 \in \mathcal{S}} \langle \mathbf{q}, \mathbf{z}(\mathbf{p}) | s_1 \rangle \pi(s_1 | s_0) \\
 &= \inf_{\mathcal{Y}} \sup_{\mathcal{V}} y_0^k \cdot \bar{k} + F[s_0, v_0] - y_0^v \cdot v_0 \\
 &\quad + \inf_{\mathbf{Q}(\mathcal{Y})} \sup_{\mathbf{P}(v_0)} -y_0^k \cdot k_0 + y_0^v \cdot W^v[s_0, a_0, M^v[s_0, v_1(\cdot)]] \\
 &\quad + m_0 \cdot H[k_0, s_0, a_0, v_1(\cdot)] + \delta \sum_{s_1 \in \mathcal{S}} \langle \mathbf{q}, \mathbf{z}(\mathbf{p}) | s_1 \rangle \pi(s_1 | s_0),
 \end{aligned} \tag{26}$$

where the second equality uses the definitions of $\mathcal{L}$, $\mathcal{Y}$, $\mathbf{Q}(\cdot)$, $\mathcal{V}$, and $\mathbf{P}(\cdot)$ and the decomposition property of inf and sup operations over product sets. The third equality uses the additive separability of $\mathcal{L}$. Once y_0^k and y_0^v are chosen, the Lagrangian is additively separable in v_0 and all other variables, including the remaining dual variables. Hence, given (y_0^k, y_0^v) , the inf over these remaining dual variables and the sup over v_0 do not affect each other and may be interchanged.

Combining (26) with the definition of D^* gives (21) in the proposition. For each $(s, y) = (s, y^k, y^v) \in \mathcal{S} \times \mathcal{Y}$,

$$\begin{aligned}
 D^*(s, y) &= \inf_{\mathbf{Q}(y)} \sup_{\mathbf{P}(v_0)} -y^k \cdot k_0 + y^v \cdot W^v[s, a_0, M^v[s, v_1(\cdot)]] \\
 &\quad + m_0 \cdot H[k_0, s, a_0, v_1] + \delta \sum_{s_1 \in \mathcal{S}} \langle \mathbf{q}, \mathbf{z}(\mathbf{p}) | s_1 \rangle \pi(s_1 | s_0) \\
 &= \inf_{\mathbf{Q}(y)} \sup_{\mathbf{P}(v_0)} -y^k \cdot k_0 + y^v \cdot W^v[s, a_0, M^v[s, v_1(\cdot)]] + m_0 \cdot H[k_0, s, a_0, v_1(\cdot)] \\
 &\quad - \delta \sum_{s_1 \in \mathcal{S}} y_1^v(s_1) \cdot v_1(s_1) \pi(s_1 | s) + \delta \sum_{s_1 \in \mathcal{S}} y_1^k(s_1) \cdot \{W^k[k_0, s, a_0] - k_1(s_1)\} \pi(s_1 | s) \\
 &\quad + \delta \sum_{s_1 \in \mathcal{S}} \left\{ y_1^v(s_1) \cdot W^v[s_1, a_1(s_1), M^v[s_1, v_2(s_1, \cdot)]] \right. \\
 &\quad \left. + m_1(s_1) \cdot H[k_1, s_1, a_1(s_1), v_2(s_1, \cdot)] + \delta \sum_{s_2 \in \mathcal{S}} \langle \mathbf{q}, \mathbf{z}(\mathbf{p}) | s_1, s_2 \rangle \pi(s_2 | s_1) \right\} \pi(s_1 | s),
 \end{aligned}$$

where the second equality expands $\langle \mathbf{q}, \mathbf{z}(\mathbf{p}) | s_1 \rangle$. Once $q_0 = (m_0, y_1^k, y_1^v)$ is chosen, $p_0 = (k_0, a_0, v_1)$ is independent of the remaining dual variables. Hence, given q_0 , the inf over these dual variables and the sup over p_0 may be interchanged to give

$$\begin{aligned}
 D^*(s, y) = & \inf_{\mathcal{Q}} \sup_{\mathcal{P}} -y_0^k \cdot k_0 + y_0^v \cdot W^v[s, a_0, M^v[s, v_1(\cdot)]] + m_0 \cdot H[k_0, s, a_0, v_1(\cdot)] \\
 & - \delta \sum_{s_1 \in \mathcal{S}} y_1^v(s_1) v_1(s_1) \pi(s_1 | s) + \sum_{s_1 \in \mathcal{S}} y_1^k(s_1) \cdot W^k[k_0, s, a_0] \pi(s_1 | s) \\
 & + \delta \inf_{\tilde{\mathcal{Q}}(q_0)} \sup_{\tilde{\mathcal{P}}(p_0)} \sum_{s_1 \in \mathcal{S}} \left\{ -y_1^k(s_1) \cdot k_1(s_1) + y_1^v(s_1) \cdot W^v[s_1, a_1(s_1), M^v[s_1, v_2(s_1, \cdot)]] \right. \\
 & \left. + m_1(s_1) \cdot H[k_1, s_1, a_1(s_1), v_2(s_1, \cdot)] + \delta \sum_{s_2 \in \mathcal{S}} \langle \mathbf{q}, \mathbf{z}(\mathbf{p}) | s_1, s_2 \rangle \pi(s_2 | s_1) \right\} \pi(s_1 | s),
 \end{aligned} \tag{27}$$

where $\tilde{\mathcal{Q}}(q_0)$ denotes the continuation of dual plans in $\mathcal{Q}$ after the exclusion of q_0 and $\tilde{\mathcal{P}}(p_0)$ denotes the continuation of primal plans in $\mathcal{P}$ after the exclusion of p_0 . Then,

$$\begin{aligned}
 D^*(s, y) = & \inf_{\mathcal{Q}} \sup_{\mathcal{P}} -y_0^k \cdot k_0 + y_0^v \cdot W^v[s, a_0, M^v[s, v_1(\cdot)]] + m_0 \cdot H[k_0, s, a_0, v_1(\cdot)] \\
 & - \delta \sum_{s_1 \in \mathcal{S}} y_1^v(s_1) v_1(s_1) \pi(s_1 | s) + \sum_{s_1 \in \mathcal{S}} y_1^k(s_1) \cdot W^k[k_0, s, a_0] \pi(s_1 | s) \\
 & + \delta \sum_{s_1 \in \mathcal{S}} \left\{ \inf_{\mathcal{Q}(y_1(s_1))} \sup_{\mathcal{P}(v_1(s_1))} -y_1^k(s_1) \cdot k_1(s_1) \right. \\
 & + y_1^v(s_1) \cdot W^v[s_1, a_1(s_1), M^v[s_1, v_2(s_1, \cdot)]] \\
 & + m_1(s_1) \cdot H[k_1, s_1, a_1(s_1), v_2(s_1, \cdot)] \\
 & \left. + \delta \sum_{s_2 \in \mathcal{S}} \langle \mathbf{q}, \mathbf{z}(\mathbf{p}) | s_1, s_2 \rangle \pi(s_2 | s_1) \right\} \pi(s_1 | s) \\
 = & \inf_{\mathcal{Q}} \sup_{\mathcal{P}} -y_0^k \cdot k_0 + y_0^v \cdot W^v[s, a_0, M^v[s, v_1(\cdot)]] + m_0 \cdot H[k_0, s, a_0, v_1(\cdot)] \\
 & - \delta \sum_{s_1 \in \mathcal{S}} y_1^v(s_1) \cdot v_1(s_1) \pi(s_1 | s) + \delta \sum_{s_1 \in \mathcal{S}} y_1^k(s_1) \cdot W^k[k_0, s, a_0] \pi(s_1 | s) \\
 & + \delta \sum_{s_1 \in \mathcal{S}} D^*(s_1, y_1(s_1)) \pi(s_1 | s),
 \end{aligned} \tag{28}$$

where the first equality uses (27), the additive separability in s_1 of the terms in the last row of (27), and the fact that, following the argument in Proposition 2, each inf-sup in the curly brackets of (28) is bounded above and, hence, the inf-sup of the sum equals the sum of the inf-sup's (although some may equal $-\infty$). The second equality then follows from the definition of D^* . Combining the last equality with the definition of J gives the second equality (22) in the proposition. *Q.E.D.*

PROOF OF PROPOSITION 4: (*Only if*) Let $J_0(y_0, v_0) = F[s_0, v_0] - y_0^v \cdot v_0 + y_0^k \cdot \bar{k}$. Then from the definition of the Lagrangian for all $(\mathbf{p}, \mathbf{q}) \in \mathbf{P} \times \mathbf{Q}$,

$$\begin{aligned} \mathcal{L}(\mathbf{p}, \mathbf{q}) &= J_0(y_0, v_0) - y_0^k \cdot k_0 + y_0^v \cdot W^v[s_0, a_0, M^v[s_0, v_1(\cdot)]] + m_0 \cdot H[k, s_0, a_0, v_1(\cdot)] \\ &\quad + \delta \sum_{s_1 \in \mathcal{S}} \langle \mathbf{q}, \mathbf{z}(\mathbf{p})|_{s_1} \rangle \pi(s_1|s_0). \end{aligned}$$

To show $D_0^* \geq J_0(y_0^*, v_0^*) + D^*(s_0, y_0^*)$, note that by definition, $D_0^* = \mathcal{L}(\mathbf{p}^*, \mathbf{q}^*)$ and

$$\begin{aligned} \mathcal{L}(\mathbf{p}^*, \mathbf{q}^*) &= \sup_{\mathbf{p}} J_0(y_0^*, v_0) - y_0^{k*} \cdot k_0 + y_0^{v*} \cdot W^v[s_0, a_0, M^v[s_0, v_1(\cdot)]] \\ &\quad + m_0^* \cdot H[k_0, s_0, a_0, v_1(\cdot)] + \delta \sum_{s_1 \in \mathcal{S}} \langle \mathbf{q}^*, \mathbf{z}(\mathbf{p})|_{s_1} \rangle \pi(s_1|s_0) \\ &= \sup_{\mathcal{V}} J_0(y_0^*, v_0) + \sup_{\mathbf{P}(v_0)} -y_0^{k*} \cdot k_0 + y_0^{v*} \cdot W^v[s_0, a_0, M^v[s_0, v_1(\cdot)]] \\ &\quad + m_0^* \cdot H[k_0, s_0, a_0, v_1(\cdot)] + \delta \sum_{s_1 \in \mathcal{S}} \langle \mathbf{q}^*, \mathbf{z}(\mathbf{p})|_{s_1} \rangle \pi(s_1|s_0) \tag{29} \\ &\geq J_0(y_0^*, v_0^*) + \inf_{\mathbf{Q}(y_0^*)} \sup_{\mathbf{P}(v_0^*)} -y_0^{k*} \cdot k_0 + y_0^{v*} \cdot W^v[s_0, a_0, M^v[s_0, v_1(\cdot)]] \\ &\quad + m_0 \cdot H[k_0, s_0, a_0, v_1(\cdot)] + \delta \sum_{s_1 \in \mathcal{S}} \langle \mathbf{q}, \mathbf{z}(\mathbf{p})|_{s_1} \rangle \pi(s_1|s_0) \\ &= J_0(y_0^*, v_0^*) + D^*(s_0, y_0^*), \end{aligned}$$

where the first equality uses the definition of $\mathbf{p}^*$, the second uses the decomposition and moving forward over constants properties of supremum operations, the inequality uses the feasibility of v_0^* for the supremum over $\mathcal{V}$ and the feasibility of the continuation of $\mathbf{q}^*$ for the continuation minimization described in this line, and the final equality uses the definition of D^* . For the reverse inequality, note first that

$$\begin{aligned} \mathcal{L}(\mathbf{p}^*, \mathbf{q}^*) &= \inf_{\mathbf{Q}} \sup_{\mathbf{p}} J_0(y_0, v_0) - y_0^k \cdot k_0 + y_0^v \cdot W^v[s_0, a_0, M^v[s_0, v_1(\cdot)]] \\ &\quad + m_0 \cdot H[k_0, s_0, a_0, v_1(\cdot)] + \delta \sum_{s_1 \in \mathcal{S}} \langle \mathbf{q}, \mathbf{z}(\mathbf{p})|_{s_1} \rangle \pi(s_1|s_0) \\ &\leq \sup_{\mathbf{p}} J_0(y_0^*, v_0) - y_0^{k*} \cdot k_0 + y_0^{v*} \cdot W^v[s_0, a_0, M^v[s_0, v_1(\cdot)]] \\ &\quad + m_0 \cdot H[k_0, s_0, a_0, v_1(\cdot)] + \delta \sum_{s_1 \in \mathcal{S}} \langle \mathbf{q}, \mathbf{z}(\mathbf{p})|_{s_1} \rangle \pi(s_1|s_0) \quad \forall \mathbf{q} \in \mathbf{Q}(y_0^*) \tag{30} \\ &= J_0(y_0^*, v_0^*) + \sup_{\mathbf{P}(v_0^*)} -y_0^{k*} \cdot k_0 + y_0^{v*} \cdot W^v[s_0, a_0, M^v[s_0, v_1(\cdot)]] \\ &\quad + m_0^* \cdot H[k_0, s_0, a_0, v_1(\cdot)] + \delta \sum_{s_1 \in \mathcal{S}} \langle \mathbf{q}, \mathbf{z}(\mathbf{p})|_{s_1} \rangle \pi(s_1|s_0), \quad \forall \mathbf{q} \in \mathbf{Q}(y_0^*), \end{aligned}$$

where the first inequality uses the feasibility of $\mathbf{q} \in \mathbf{Q}(y_0^*)$ for the inf problems, the second equality uses the decomposition property of the sup operator and the maximality of v_0^* in

$\sup_v J_0^*(y_0^*, v_0)$. Since the last equality in (30) holds $\forall \mathbf{q} \in \mathbf{Q}(y_0^*)$:

$$\begin{aligned} D_0^* &\leq J_0(y_0^*, v_0^*) + \inf_{\mathbf{Q}(y_0^*)} \sup_{\mathbf{P}(v_0)} -y_0^{k*} \cdot k_0 + y_0^{v*} \cdot W^v[s_0, a_0, M^v[s_0, v_1(\cdot)]] \\ &\quad + m_0^* \cdot H[k_0, s_0, a_0, v_1(\cdot)] + \delta \sum_{s_1 \in \mathcal{S}} \langle \mathbf{q}, \mathbf{z}(\mathbf{p}) | s_1 \rangle \pi(s_1 | s_0) \\ &= J_0(y_0^*, v_0^*) + D^*(s_0, y_0^*), \end{aligned} \quad (31)$$

where the last equality uses the definition of D^* . Combining (29) and (31) gives $D_0^* = J_0(y_0^*, v_0^*) + D^*(s_0, y_0^*)$. Thus, y_0^* attains the minimum in (21), and since v_0^* attains the maximum in $\sup_v J_0(y_0^*, v_0)$, $(y_0^*, v_0^*) \in G_0$. Repeating the argument at successive histories implies that $q_t^*(s^t) = (m_t^*(s^t), y_{t+1}^*(s^t))$ attains the minimum in (22) at $(s_t, y_t^*(s^t))$ and so $(q_t^*(s^t), p_t^*(s^t)) \in G(s_t, y_t^*(s^t))$.

(If) The remainder of the proof uses the following lemma.

LEMMA 5: Let $J_0(y_0, v_0) = F[s_0, v_0] - y_0^v \cdot v_0 + y_0^k \cdot \bar{k}$. For all $(\mathbf{p}, \mathbf{q}) \in \mathbf{P} \times \mathbf{Q}$, $\mathcal{L}(\mathbf{p}, \mathbf{q}) = J_0(y_0, v_0) + \sum_{t=0}^{\infty} \delta^t \sum_{s^t} J(s_t, y_t(s^t); q_t(s^t), p_t(s^t)) \pi^t(s^t | s_0)$.

PROOF: Expanding $\langle \mathbf{q}, \mathbf{z}(\mathbf{p}) \rangle$, the Lagrangian $\mathcal{L}$ is

$$\begin{aligned} \mathcal{L}(\mathbf{p}, \mathbf{q}) &= F[s_0, v_0] + \sum_{t=0}^{\infty} \sum_{s^t} \delta^t \{ m_t(s^t) \cdot z_t^h(\mathbf{p})(s^t) \\ &\quad + y_t^k(s^t) \cdot z_t^k(\mathbf{p})(s^t) + y_t^v(s^t) \cdot z_t^v(\mathbf{p})(s^t) \} \pi^t(s^t | s_0). \end{aligned}$$

From the definition of J , we have, for each T ,

$$\begin{aligned} \mathcal{L}(\mathbf{p}, \mathbf{q}) &= J_0(y_0, v_0) + \sum_{t=0}^T \delta^t \sum_{s^t} J(s_t, y_t(s^t); q_t(s^t), p_t(s^t)) \pi^t(s^t | s_0) \\ &\quad + \delta^T \sum_{s^T} [y_T^v(s^T) v_T(s^T) + m_T(s^T) H[k_T(s^T), s_T, v_{T+1}(s^T, \cdot)]] \\ &\quad \times \pi^T(s^T | s_0) \\ &\quad - \delta^T \sum_{s^T} y_T^k(s^T) k_T(s^T) \pi^T(s^T | s_0) + \delta^{T+1} \sum_{s^{T+1}} \langle \mathbf{q}, \mathbf{z}(\mathbf{p}) | s^T \rangle \pi^{T+1}(s^{T+1} | s_0). \end{aligned} \quad (32)$$

Since $\mathbf{q} \in \mathbf{Q}$ and since $\mathbf{v}, \mathbf{k}$, and $\mathbf{z}(\mathbf{p})$ belong to ℓ^∞ , the infinite sums in the second and third rows of (32) converge to zero as $T \rightarrow \infty$, giving the result. Q.E.D.

Let $(\mathbf{p}^*, \mathbf{q}^*)$ satisfy (i) $(y_0^{k*}, y_0^{v*}, v_0^*) \in G_0$, (ii) $\forall t \geq 1, s^t \in \mathcal{S}^t, (m_t^*(s^t), y_{t+1}^*(s^t), p_t^*(s^t)) \in G(s_t, y_t^*(s^t))$, and (iii) $\mathbf{q}^* \in \mathbf{Q}$. From the definitions of G_0 and G , $J_0(y_0^*, v_0^*) = \sup_v J_0(y_0^*, v_0)$ and $J(s_t, y_t^*(s^t); q_t^*(s^t), p_t^*(s^t)) = \sup_{\mathbf{p}} J(s_t, y_t^*(s^t); q_t^*(s^t), p_t(s^t))$. So, by

Lemma 5, $\forall \mathbf{p} \in \mathbf{P}$,

$$\begin{aligned} \mathcal{L}(\mathbf{p}^*, \mathbf{q}^*) &= J_0(y_0^*, v_0^*) + \sum_{t=0}^{\infty} \delta^t \sum_{S^t} J(s_t, y_t^*(s^t); q_t^*(s^t), p_t^*(s^t)) \pi^t(s^t | s_0) \\ &\geq J_0(y_0^*, v_0) + \sum_{t=0}^{\infty} \delta^t \sum_{S^t} J(s_t, y_t^*(s^t); q_t^*(s^t), p_t(s^t)) \pi^t(s^t | s_0), \end{aligned}$$

and $\mathbf{p}^* \in \arg\max_{\mathbf{p}} \mathcal{L}(\mathbf{p}, \mathbf{q}^*)$. Let $J_0^*(y) = \sup_{\mathbf{p}} J_0(y, v_0)$ and let $J^*(s_t, y_t(s^t); q_t(s^t)) = \sup_{\mathbf{p}} J(s_t, y_t(s^t); q_t(s^t), p)$. For all $\mathbf{q} \in \mathbf{Q}$, $\mathbf{p} \in \mathbf{P}$: $J_0^*(y_0) + \sum_{t=0}^{\infty} \delta^t \sum_{S^t} J^*(s_t, y_t(s^t); q_t(s^t)) \times \pi^t(s^t | s_0) \geq J_0(y_0, v_0) + \sum_{t=0}^{\infty} \delta^t \sum_{S^t} J(s_t, y_t(s^t); q_t(s^t), p_t(s^t)) \pi^t(s^t | s_0)$. Hence,

$$\begin{aligned} &J_0^*(y_0) + \sum_{t=0}^{\infty} \delta^t \sum_{S^t} J^*(s_t, y_t(s^t); q_t(s^t)) \pi^t(s^t | s_0) \\ &\geq \sup_{\mathbf{p}} J_0(y_0, v_0) + \sum_{t=0}^{\infty} \delta^t \sum_{S^t} J(s_t, y_t(s^t); q_t(s^t), p_t(s^t)) \pi^t(s^t | s_0). \end{aligned} \tag{33}$$

On the other hand, for small $\varepsilon > 0$, let $\hat{\mathbf{p}} \in \mathbf{P}$ be such that $J_0(y_0, \hat{v}_0) > J^*(y_0) - \frac{\varepsilon}{2}$ and for each t and s^t , $J(s_t, y_t(s^t); q_t(s^t), \hat{p}_t(s^t)) > J^*(s_t, y_t(s^t); q_t(s^t)) - (1 - \delta)\frac{\varepsilon}{2}$, then $J_0^*(y_0) + \sum_{t=0}^{\infty} \delta^t \sum_{S^t} J^*(s_t, y_t(s^t); q_t(s^t)) \pi^t(s^t | s_0) < J_0(y_0, \hat{v}_0) + \sum_{t=0}^{\infty} \delta^t \sum_{S^t} J(s_t, y_t(s^t); q_t(s^t), \hat{p}_t(s^t)) \pi^t(s^t | s_0) + \varepsilon$. Since $\varepsilon > 0$ is arbitrary, it follows that

$$\begin{aligned} &J_0^*(y_0) + \sum_{t=0}^{\infty} \delta^t \sum_{S^t} J^*(s_t, y_t(s^t); q_t(s^t)) \pi^t(s^t | s_0) \\ &\leq \sup_{\mathbf{p}} J_0(y_0, v_0) + \sum_{t=0}^{\infty} \delta^t \sum_{S^t} J(s_t, y_t(s^t); q_t(s^t), p_t(s^t)) \pi^t(s^t | s_0). \end{aligned} \tag{34}$$

Combining the definition of D_0^* , Lemma 5, and (33)–(34) and using $\mathbf{q}^* \in \mathbf{Q}$ gives

$$\begin{aligned} D_0^* &= \inf_{\mathbf{Q}} \sup_{\mathbf{P}} \mathcal{L}(\mathbf{p}, \mathbf{q}) \\ &= \inf_{\mathbf{Q}} \sup_{\mathbf{P}} J_0(y_0, v_0) + \sum_{t=0}^{\infty} \delta^t \sum_{S^t} J(s_t, y_t(s^t); q_t(s^t), p_t(s^t)) \pi^t(s^t | s_0) \\ &= \inf_{\mathbf{Q}} J_0^*(y_0) + \sum_{t=0}^{\infty} \delta^t \sum_{S^t} J^*(s_t, y_t(s^t); q_t(s^t)) \pi^t(s^t | s_0) \\ &\leq J_0^*(y_0^*) + \sum_{t=0}^{\infty} \delta^t \sum_{S^t} J^*(s_t, y_t^*(s^t); q_t^*(s^t)) \pi^t(s^t | s_0). \end{aligned} \tag{35}$$

The definitions of G_0 and G and the dual Bellman equation imply $D_0^* = J_0^*(y_0^*) + \sum_{t=0}^T \delta^t \sum_{S^t} J^*(s_t, y_t^*(s^t); q_t^*(s^t)) \pi^t(s^t | s_0) + \delta^{T+1} \sum_{S^{T+1}} D^*(s_{T+1}, y_{T+1}^*(s^{T+1})) \pi(s^{T+1} | s_0)$.

Taking the limit as T goes to infinity and using condition (T) implies

$$D_0^* \geq J_0^*(y_0^*) + \sum_{t=0}^{\infty} \delta^t \sum_{S^t} J^*(s_t, y_t^*(s^t); q_t^*(s^t)) \pi^t(s^t | s_0). \quad (36)$$

Combining (35) with (36) implies that $\mathbf{q}^*$ attains the minimum, as required. *Q.E.D.*

APPENDIX B: PROOFS FOR SECTION 5

PROOF OF LEMMA 2: Let Ψ , Φ , and Ω denote vector spaces and let $L : \Psi \times \Phi \times \Omega \rightarrow \mathbb{R} \cup \{-\infty\}$. Assume that for each $\omega \in \Omega$, $L(\cdot; \cdot, \omega)$ is sub-linear. For $\psi \in \Psi$, let $\mathcal{T}(\psi) = \inf_{\Phi} \sup_{\Omega} L(\psi; \phi, \omega)$. Assume that $\mathcal{T} < \infty$. We prove that $\mathcal{T}$ is sub-linear. We first show that $\mathcal{T}$ is convex. Let ψ^1 and ψ^2 be elements of Ψ and $\lambda \in [0, 1]$. Let $\psi^\lambda = \lambda\psi^1 + (1 - \lambda)\psi^2$. For $i = 1, 2$, let $\{\phi_n^i\}$ denote sequences such that $\mathcal{T}(\psi^i) = \inf_{\Phi} \sup_{\Omega} L(\psi^i; \phi, \omega) = \lim_{n \rightarrow \infty} \sup_{\Omega} L(\psi^i; \phi_n^i, \omega)$. Also let $\phi_n^\lambda = \lambda\phi_n^1 + (1 - \lambda)\phi_n^2$. Then,

$$\begin{aligned} \lambda \mathcal{T}(\psi^1) + (1 - \lambda) \mathcal{T}(\psi^2) &= \lambda \inf_{\Phi} \sup_{\Omega} L(\psi^1; \phi, \omega) + (1 - \lambda) \inf_{\Phi} \sup_{\Omega} L(\psi^2; \phi, \omega) \\ &= \lambda \lim_{n \rightarrow \infty} \sup_{\Omega} L(\psi^1; \phi_n^1, \omega) + (1 - \lambda) \lim_{n \rightarrow \infty} \sup_{\Omega} L(\psi^2; \phi_n^2, \omega) \\ &\geq \lim_{n \rightarrow \infty} \sup_{\Omega} \{ \lambda L(\psi^1; \phi_n^1, \omega) + (1 - \lambda) L(\psi^2; \phi_n^2, \omega) \} \\ &\geq \lim_{n \rightarrow \infty} \sup_{\Omega} L(\psi^\lambda; \phi_n^\lambda, \omega) \geq \inf_{\Phi} \sup_{\Omega} L(\psi^\lambda; \phi, \omega) = \mathcal{T}(\psi^\lambda), \end{aligned}$$

where the sum on the left-hand side of the first line is well defined because $\mathcal{T} < \infty$, the first equality follows from the definition of $\mathcal{T}$, the second equality uses the definition of the sequences $\{\phi_n^i\}$, $i = 1, 2$, the first inequality uses the fact that the sup of a sum is less than or equal to the sum of sups, and the second inequality uses the convexity of $L(\cdot; \cdot, \omega)$. Thus, $\mathcal{T}$ is convex. Next, we show positive homogeneity. Suppose that $\psi \in \Psi$ and $\lambda > 0$. Then $\mathcal{T}(\lambda\psi) = \inf_{\Phi} \sup_{\Omega} L(\lambda\psi; \phi, \omega) = \lambda \inf_{\Phi} \sup_{\Omega} L(\psi; \phi/\lambda, \omega) = \lambda \mathcal{T}(\psi)$, where the second equality uses the positive homogeneity of $L(\cdot, \cdot, \omega)$. Thus, $\mathcal{T}$ is positively homogeneous of degree 1 and, combining results, sub-linear.

(i) For fixed $s_0 \in \mathcal{S}$ and a pair $(\mathbf{p}, \mathbf{q}) \in \mathbf{P} \times \mathbf{Q}$, let

$$\begin{aligned} L(y_0; (m_0, \{\mathbf{q}|s_1\}), (p, \{\mathbf{z}(\mathbf{p})|s_1\})) &:= -y_0^k \cdot k_0 + y_0^v \cdot W^v[s_0, a_0, M^v[s_0, v_1]] \\ &\quad + m_0 \cdot H[k_0, s_0, a_0, v_1] + \delta \sum_{s_1 \in \mathcal{S}} \langle \mathbf{q}, \mathbf{z}(\mathbf{p})|s_1 \rangle \pi(s_1 | s_0). \end{aligned}$$

Then, $D^*(s_0, y_0) = \inf_{\mathbf{Q}(\mathbf{y}_0)} \sup_{\mathbf{P}(\mathbf{y}_0)} L(y_0; (m_0, \{\mathbf{q}|s_1\}), (p, \{\mathbf{z}(\mathbf{p})|s_1\}))$. By Proposition 2, $D^*(s_0, \cdot) < \infty$. Also, for each $\omega = (p, \{\mathbf{z}(\mathbf{p})|s_1\})$, the function $L(\cdot; \cdot, \omega)$ is linear and, hence, sub-linear. Applying the general result from the first part of the proof, $D^*(s_0, \cdot)$ is sub-linear. Since s_0 was arbitrary in $\mathcal{S}$, D^* is sub-linear. (ii) Recall that

$$\mathcal{B}(D)(s, y) = \inf_{\mathbf{Q}} \sup_{\mathbf{P}} J(s, y; q, p) + \delta \sum_{s' \in \mathcal{S}} D(s', y'(s')) \pi(s' | s).$$

For $q = 0$, $\sup_{\mathbf{P}} J(s, y; 0, p) + \delta \sum_{s' \in \mathcal{S}} D(s', 0) \pi(s' | s) < \infty$. Hence, $\mathcal{B}(D)(s, y) < \infty$. Also, for each (s, p) , $J(s, \cdot; \cdot, p)$ is linear and D is sub-linear by assumption. Thus, the function

$J(s, y; q, p) + \delta \sum_{s' \in \mathcal{S}} D(s', y'(s')) \pi(s'|s)$ is sub-linear. Thus, again by the general result, $\mathcal{B}(D)(s, \cdot)$ is sub-linear and, hence, $\mathcal{B}(D)$ is sub-linear. *Q.E.D.*

PROOF OF LEMMA 3: Evidently, $(\mathcal{G}, d)$ is a metric space. Let $\{D_n\}$ be a Cauchy sequence in $\mathcal{G}$. Thus, as $n, m \rightarrow \infty$,

$$d(D_n, D_m) = \sup_{\mathcal{S} \times \mathcal{C}} \left| \ln \left(\frac{D_n(s, y) - \underline{D}(s, y)}{\underline{D}(s, y) - \underline{\underline{D}}(s, y)} \right) - \ln \left(\frac{D_m(s, y) - \underline{D}(s, y)}{\underline{D}(s, y) - \underline{\underline{D}}(s, y)} \right) \right| \rightarrow 0.$$

For each $n = 1, 2, \dots$, define $g_n : \mathcal{S} \times \mathcal{C} \rightarrow \mathbb{R}$ according to: $g_n(s, y) = \ln \left(\frac{D_n(s, y) - \underline{D}(s, y)}{\underline{D}(s, y) - \underline{\underline{D}}(s, y)} \right)$, $(s, y) \in \mathcal{S} \times \mathcal{C}$. Let $\underline{g} = 0$ and $\bar{g} = \ln \left(\frac{\bar{D} - \underline{D}}{\underline{D} - \underline{\underline{D}}} \right) \geq \ln \left(\frac{\bar{D} - \underline{D}}{\underline{D} - \underline{\underline{D}}} \right)$. Note that $\bar{g}$ is bounded in $\mathcal{S} \times \mathcal{C}$ by the continuity of each $\bar{D}(s, \cdot) - \underline{D}(s, \cdot)$ and the compactness of $\mathcal{C}$. The sequence of functions $\{g_n\}$ is Cauchy with respect to the sup-norm and for each n , $\underline{g} \leq g_n \leq \bar{g}$. By the completeness (with respect to the sup-norm) of the bounded functions from $\mathcal{C}$ to $\mathbb{R}$, $\{g_n\}$ converges in the sup-norm to a function g_∞ , with each $g_\infty(s, \cdot)$ bounded and $\underline{g} \leq g_\infty \leq \bar{g}$. Use g_∞ to define the homogeneous function D_∞ as

$$D_\infty(s, y) = \|y\|_E \left\{ \underline{D} \left(s, \frac{y}{\|y\|_E} \right) + \exp \left\{ g_\infty \left(s, \frac{y}{\|y\|_E} \right) \right\} \left(\underline{D} \left(s, \frac{y}{\|y\|_E} \right) - \underline{\underline{D}} \left(s, \frac{y}{\|y\|_E} \right) \right) \right\}.$$

By construction, $\underline{D} \leq D_\infty \leq \bar{D}$ and $D_n \xrightarrow{d} D_\infty$. Since D_∞ is the pointwise limit of a sequence of sub-linear and, so, convex functions, it is convex. Thus, it is in $\mathcal{G}$. *Q.E.D.*

PROOF OF PROPOSITION 5: Let $\mathcal{G}_0$ denote the interval of real-valued functions between $\underline{D}$ and $\bar{D}$. Let D_1 and D_2 be any pair of functions in $\mathcal{G}_0$ and let $\lambda \in [0, 1]$. Define, for each $(s, y, q) \in \mathcal{S} \times \mathcal{Y} \times \mathcal{Q}$, $J^*(s, y, q) = \sup_p J(s, y; q, p) \in \mathbb{R}$. Then, for each $(s, y) \in \mathcal{S} \times \mathcal{Y}$,

$$\begin{aligned} & \mathcal{B}(\lambda D_1 + (1 - \lambda) D_2)(s, y) \\ &= \inf_{\mathcal{Q}} J^*(s, y, m, y') + \delta \sum_{s' \in \mathcal{S}} \{ \lambda D_1(s', y'(s')) + (1 - \lambda) D_2(s', y'(s')) \} \pi(s'|s) \\ &= \inf_{\mathcal{Q}} \left\{ J^*(s, y, m, y') + \delta \sum_{s' \in \mathcal{S}} D_1(s', y'(s')) \pi(s'|s) \right\} \\ & \quad + (1 - \lambda) \left\{ J^*(s, y, m, y') + \delta \sum_{s' \in \mathcal{S}} D_2(s', y'(s')) \pi(s'|s) \right\} \\ &\geq \lambda \inf_{\mathcal{Q}} \left\{ J^*(s, y, m, y') + \delta \sum_{s' \in \mathcal{S}} D_1(s', y'(s')) \pi(s'|s) \right\} \\ & \quad + (1 - \lambda) \inf_{\mathcal{Q}} \left\{ J^*(s, y, m, y') + \delta \sum_{s' \in \mathcal{S}} D_2(s', y'(s')) \pi(s'|s) \right\} \\ &= \lambda \mathcal{B}(D_1)(s, y) + (1 - \lambda) \mathcal{B}(D_2)(s, y), \end{aligned}$$

where, recall, $\mathcal{B}(D) < \infty$; hence, the sums in the last two rows are well defined. Thus, $\mathcal{B}$ is concave on $\mathcal{G}_0$. Let $D_1, D_2 \in \mathcal{G} \subset \mathcal{G}_0$. By definition of d , for each $(s, y) \in \mathcal{S} \times \mathcal{C}$,

$$\ln\left(\frac{D_2(s, y) - \underline{\underline{D}}(s, y)}{\underline{D} - \underline{\underline{D}}}\right) \leq \ln\left(\frac{D_1(s, y) - \underline{\underline{D}}(s, y)}{\underline{D}(s, y) - \underline{\underline{D}}(s, y)}\right) + d(D_1, D_2).$$

Taking the exponential of each side and rearranging gives

$$\exp\{-d(D_1, D_2)\} \left(\frac{D_2(s, y) - \underline{\underline{D}}(s, y)}{\underline{D}(s, y) - \underline{\underline{D}}(s, y)}\right) \leq \left(\frac{D_1(s, y) - \underline{\underline{D}}(s, y)}{\underline{D}(s, y) - \underline{\underline{D}}(s, y)}\right).$$

But, by Assumption 3(i), $\underline{D} - \underline{\underline{D}} > 0$ and so, after rearrangement,

$$D_1(s, y) \geq \exp\{-d(D_1, D_2)\} D_2(s, y) + (1 - \exp\{-d(D_1, D_2)\}) \underline{\underline{D}}(s, y).$$

Since D_1, D_2 , and $\underline{\underline{D}}$ are positively homogeneous of degree 1, this inequality holds at all $(s, y) \in \mathcal{S} \times \mathcal{Y}$. Then, by monotonicity and concavity of $\mathcal{B}$ (on $\mathcal{G}_0$),

$$\begin{aligned} \mathcal{B}(D_1) &\geq \mathcal{B}(\exp\{-d(D_1, D_2)\} D_2 + (1 - \exp\{-d(D_1, D_2)\}) \underline{\underline{D}}) \\ &\geq \exp\{-d(D_1, D_2)\} \mathcal{B}(D_2) + (1 - \exp\{-d(D_1, D_2)\}) \mathcal{B}(\underline{\underline{D}}). \end{aligned} \quad (\text{D1})$$

By assumption, there is an $\varepsilon > 0$ such that, for each $(s, y) \in \mathcal{S} \times \mathcal{C}$, $\mathcal{B}(\underline{\underline{D}})(s, y) > \underline{\underline{D}}(s, y) + \varepsilon$. For $(s, y) \in \mathcal{S} \times \mathcal{C}$, define

$$\lambda(s, y) := \frac{\varepsilon}{\overline{D}(s, y) - \underline{\underline{D}}(s, y)}.$$

Since $\overline{D}(s, y) \geq \mathcal{B}(\overline{D})(s, y) \geq \mathcal{B}(\underline{\underline{D}})(s, y) > \underline{\underline{D}}(s, y) + \varepsilon$, $\lambda(s, y) \in (0, 1)$. Now, for each $s \in \mathcal{S}$, $\underline{\underline{D}}(s, \cdot)$ and $\overline{D}(s, \cdot)$ are continuous. Thus, $\lambda(s, \cdot)$ is continuous, and since $\mathcal{C}$ is compact, there is a $\lambda^* = \min_{\mathcal{S} \times \mathcal{C}} \lambda(s, y) \in (0, 1)$. Then, for all $(s, y) \in \mathcal{S} \times \mathcal{C}$,

$$\begin{aligned} \mathcal{B}(\underline{\underline{D}})(s, y) &> \underline{\underline{D}}(s, y) + \varepsilon = \lambda(s, y) \overline{D}(s, y) + (1 - \lambda(s, y)) \underline{\underline{D}}(s, y) \\ &\geq \lambda^* \overline{D}(s, y) + (1 - \lambda^*) \underline{\underline{D}}(s, y) \\ &\geq \lambda^* \mathcal{B}(D_2)(s, y) + (1 - \lambda^*) \underline{\underline{D}}(s, y), \end{aligned} \quad (\text{D2})$$

where the first inequality is by assumption, the first equality uses the definition of $\lambda(s, y)$, the second inequality uses the definition of λ^* and $\overline{D} \geq \underline{\underline{D}}$, and the final inequality uses $\overline{D} \geq \mathcal{B}(\overline{D}) \geq \mathcal{B}(D_2)$. Combining (B) with (B) gives, for all $(s, y) \in \mathcal{C}$,

$$\begin{aligned} \mathcal{B}(D_1)(s, y) &\geq \exp\{-d(D_1, D_2)\} \mathcal{B}(D_2)(s, y) + (1 - \exp\{-d(D_1, D_2)\}) \\ &\quad \times [\lambda^* \mathcal{B}(D_2)(s, y) + (1 - \lambda^*) \underline{\underline{D}}(s, y)]. \end{aligned}$$

Letting $r := \exp\{-d(D_1, D_2)\} + (1 - \exp\{-d(D_1, D_2)\}) \lambda^*$ then gives, for $(s, y) \in \mathcal{S} \times \mathcal{C}$,

$$\frac{\mathcal{B}(D_1)(s, y) - \underline{\underline{D}}(s, y)}{\underline{D}(s, y) - \underline{\underline{D}}(s, y)} \geq r \frac{\mathcal{B}(D_2)(s, y) - \underline{\underline{D}}(s, y)}{\underline{D}(s, y) - \underline{\underline{D}}(s, y)}.$$

Hence, taking logs, for $(s, y) \in \mathcal{S} \times \mathcal{C}$,

$$\ln\left(\frac{\mathcal{B}(D_1)(s, y) - \underline{\underline{D}}(s, y)}{\underline{D}(s, y) - \underline{\underline{D}}(s, y)}\right) \geq \ln r + \ln\left(\frac{\mathcal{B}(D_2)(s, y) - \underline{\underline{D}}(s, y)}{\underline{D}(s, y) - \underline{\underline{D}}(s, y)}\right).$$

But from the definition of r and Jensen's inequality,

$$\ln r \geq (1 - \lambda^*) \ln \exp\{-d(D_1, D_2)\} + \lambda^* \ln 1 = -(1 - \lambda^*)d(D_1, D_2).$$

Thus, for $(s, y) \in \mathcal{S} \times \mathcal{C}$,

$$\begin{aligned} (1 - \lambda^*)d(D_1, D_2) &\geq -\ln r \\ &\geq \ln\left(\frac{\mathcal{B}(D_2)(s, y) - \underline{\underline{D}}(s, y)}{\underline{D}(s, y) - \underline{\underline{D}}(s, y)}\right) - \ln\left(\frac{\mathcal{B}(D_1)(s, y) - \underline{\underline{D}}(s, y)}{\underline{D}(s, y) - \underline{\underline{D}}(s, y)}\right). \end{aligned} \quad (D3)$$

Repeating the argument with D_1 and D_2 interchanged and combining with (B) implies that, for all $(s, y) \in \mathcal{S} \times \mathcal{C}$,

$$(1 - \lambda^*)d(D_1, D_2) \geq \left| \ln\left(\frac{\mathcal{B}(D_2)(s, y) - \underline{\underline{D}}(s, y)}{\underline{D}(s, y) - \underline{\underline{D}}(s, y)}\right) - \ln\left(\frac{\mathcal{B}(D_1)(s, y) - \underline{\underline{D}}(s, y)}{\underline{D}(s, y) - \underline{\underline{D}}(s, y)}\right) \right|.$$

Consequently, letting $\rho := (1 - \lambda^*) \in (0, 1)$,

$$\begin{aligned} \rho d(D_1, D_2) &\geq \sup_{\mathcal{S} \times \mathcal{C}} \left| \ln\left(\frac{\mathcal{B}(D_2)(s, y) - \underline{\underline{D}}(s, y)}{\underline{D}(s, y) - \underline{\underline{D}}(s, y)}\right) - \ln\left(\frac{\mathcal{B}(D_1)(s, y) - \underline{\underline{D}}(s, y)}{\underline{D}(s, y) - \underline{\underline{D}}(s, y)}\right) \right| \\ &= d(\mathcal{B}(D_1), \mathcal{B}(D_2)), \end{aligned}$$

as desired. Q.E.D.

We now show that the Bellman operator's iterates converge uniformly to D^* on $\mathcal{S} \times \mathcal{C}$. For sub-linear functions D_1, D_2 , define $\|D_1 - D_2\| := \sup_{\mathcal{S} \times \mathcal{C}} |D_1(s, y) - D_2(s, y)|$.

PROPOSITION 10: *Let Assumption 3 hold. (i) For any two functions $D_1, D_2 \in \mathcal{G}$, we have $\|D_1 - \underline{\underline{D}}\|, \|D_2 - \underline{\underline{D}}\| \leq \|\bar{D} - \underline{\underline{D}}\| < \infty$, and $\|D_1 - D_2\| \leq \|\bar{D} - \underline{\underline{D}}\|(\exp d(D_1, D_2) - 1)$. (ii) Let $\underline{D} \leq D^* \leq \bar{D}$. For any initial function $D_0 \in \mathcal{G}$, we have $\lim_{n \rightarrow \infty} \|\mathcal{B}^n(D_0) - D^*\| = 0$, that is, the dual Bellman operator's iterates converge uniformly to D^* in $\mathcal{S} \times \mathcal{C}$.*

PROOF: (i) Since $D_1, D_2 \in \mathcal{G}$, the inequalities $\|D_1 - \underline{\underline{D}}\|, \|D_2 - \underline{\underline{D}}\| \leq \|\bar{D} - \underline{\underline{D}}\|$ are immediate, the boundedness comes from the continuity of $\bar{D}$ and $\underline{\underline{D}}$ and compactness of $\mathcal{S} \times \mathcal{C}$. Now, for any two distinct functions D_1 and D_2 , we have $d(D_1, D_2) > 0$; hence, $\mu := \exp d(D_1, D_2) > 1$. From the definition of the Thompson distance, we have $\mu = \exp d(D_1, D_2) \geq \left| \frac{D_1(s, y) - \underline{\underline{D}}(s, y)}{D_2(s, y) - \underline{\underline{D}}(s, y)} \right|$ for all (s, y) . We hence have (pointwise) that both $D_1 - \underline{\underline{D}} \leq \mu(D_2 - \underline{\underline{D}})$ and $D_2 - \underline{\underline{D}} \leq \mu(D_1 - \underline{\underline{D}})$. Therefore, $D_1 - D_2 \leq (\mu - 1)(D_2 - \underline{\underline{D}})$ and $D_2 - D_1 \leq (\mu - 1)(D_1 - \underline{\underline{D}})$. This implies that (again pointwise) $-(\mu - 1)|D_2 - \underline{\underline{D}}| \leq D_2 - D_1 \leq (\mu - 1)|D_1 - \underline{\underline{D}}|$. From this, we have $|D_2 - D_1| \leq (\mu - 1) \max\{|D_2 - \underline{\underline{D}}|, |D_1 - \underline{\underline{D}}|\} \leq (\mu - 1)\|\bar{D} - \underline{\underline{D}}\|$, delivering $\|D_1 - D_2\| \leq \|\bar{D} - \underline{\underline{D}}\|(\mu - 1)$ as we stated in the proposition. (ii) Immediate from point (i) and Theorem 2 which implies $\exp d(\mathcal{B}^n(D), D^*) \rightarrow 1$. Q.E.D.

APPENDIX C: PROOFS FOR SECTION 6

PROOF OF PROPOSITION 6: Equality of values follows from standard arguments (e.g., Rockafellar (1974), Theorem 2). Saddle point existence implies that both the dual and primal problems have solutions (again see Rockafellar (1974), Theorem 2). If $\mathbf{q}^*$ solves the dual problem (i.e., $\mathbf{q}^*$ attains the minimum in (IS)) and $\mathbf{p}^*$ solves the primal problem, then from the definitions of these problems, $D_0^* = \sup_{\mathbf{p} \in \mathbf{P}} \mathcal{L}(\mathbf{p}, \mathbf{q}^*) \geq \mathcal{L}(\mathbf{p}^*, \mathbf{q}^*) \geq \inf_{\mathbf{q} \in \mathbf{Q}} \mathcal{L}(\mathbf{p}^*, \mathbf{q}) = P_0^*$. Hence, equality of values $D_0^* = P_0^*$ implies that $\mathbf{p}^*$ solves the inner supremum in the dual problem and $(\mathbf{q}^*, \mathbf{p}^*) \in \Lambda$. The desired result then follows from Proposition 4. Q.E.D.

PROOF OF PROPOSITION 7: Since a saddle point exists, the set of primal plans $\mathbf{P}^*$ that solve (P) is nonempty. From Proposition 6, whenever $\mathcal{L}$ admits a saddle, $\mathbf{P}^* \subset \hat{\mathbf{P}}$. By assumption, $\hat{\mathbf{P}}$ only contains the element $\hat{\mathbf{p}}$. Hence $\mathbf{P}^* = \{\hat{\mathbf{p}}\}$. Q.E.D.

PROOF OF PROPOSITION 8: This proof is long and is reported in Supplemental Material Appendix C. It is structured as follows. First, existence of a saddle point with summable multipliers is established for an abstract problem with inequality constraints. Next, this problem is related to a modified version of (P) (called (MP)). A Lagrangian is associated with (MP) and it is shown that each primal plan solving (MP) is part of a saddle point with a minimizing summable multiplier. Finally, it is shown that each solution to (P) defines a solution to (MP) and the minimizing multiplier from (MP) is used to construct a minimizing multiplier and, hence, saddle point for (P). Q.E.D.

PROOF OF PROPOSITION 9: If $(\hat{\mathbf{p}}, \hat{\mathbf{q}})$ is a candidate plan satisfying Condition (T), then from Proposition 4, it is a solution to the dual problem (IS). Moreover, from the dual Bellman equation (21) in Proposition 3, $(\hat{\mathbf{p}}, \hat{\mathbf{q}})$ solves $D_0^* = F[s_0, \hat{v}_0] - \hat{y}_0^v \cdot \hat{v}_0 + \hat{y}_0^k \cdot \bar{k} + D^*(s_0, \hat{y}_0^k, \hat{y}_0^v)$. Condition (ii) implies $F[s_0, \hat{v}_0] \geq D_0^*$, and hence, from the weak duality inequality, $F[s_0, \hat{v}_0] \geq P_0^*$. The feasibility condition (iii) then implies that $P_0^* \geq F[s_0, \hat{v}_0]$. Combining inequalities $F[s_0, \hat{v}_0] = P_0^*$ and $\hat{\mathbf{p}}$ solves (P). In addition, $\mathcal{L}(\hat{\mathbf{p}}, \hat{\mathbf{q}}) = \sup_{\mathbf{p}} \mathcal{L}(\mathbf{p}, \hat{\mathbf{q}}) = D_0^* = P_0^* = \inf_{\mathbf{q}} \mathcal{L}(\hat{\mathbf{p}}, \mathbf{q})$, where the first and second inequalities use the fact that $(\hat{\mathbf{p}}, \hat{\mathbf{q}})$ solves the dual, the third uses the result above that $D_0^* = P_0^*$, and the fourth the fact that $\hat{\mathbf{p}}$ solves (P) and, so, maximizes $\inf_{\mathbf{q}} \mathcal{L}(\mathbf{p}, \mathbf{q})$ and attains P_0^* . Thus, $\hat{\mathbf{p}}$ solves $\max \mathcal{L}(\mathbf{p}, \hat{\mathbf{q}})$ and $\hat{\mathbf{q}}$ solves $\min \mathcal{L}(\hat{\mathbf{p}}, \mathbf{q})$ and $(\hat{\mathbf{p}}, \hat{\mathbf{q}})$ is a saddle for $\mathcal{L}$. Q.E.D.

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